

BUNI-002: Red Team and Reconstruction of the UN Sustainable Development Goals

Executive Summary

The **United Nations Sustainable Development Goals (SDGs)** – a set of 17 global goals for 2030 – are intended as a transformative agenda. This report applies a rigorous **adversarial red team evaluation** to the SDGs, stress-testing each goal against potential manipulation, misalignment, and failure points. We identify structural weaknesses that undermine each goal's resilience under hostile conditions, and we propose reconstructed goals designed to be more robust, measurable, and fair. Our analysis yields the following high-level findings and recommendations:

- **Structural Vulnerabilities:** The current SDG framework exhibits multiple vulnerabilities – goals are often **non-binding, underfunded, vague, and susceptible to political capture** ¹ ² . Several goals contain internal contradictions (e.g. economic growth vs. environmental limits) and rely on metrics that can be easily gamed, jeopardizing their integrity under adversarial scrutiny ³ ⁴ .
- **Five-Axis Red Team Assessment:** We evaluated each SDG along five adversarial dimensions – *Symmetry Integrity, Incentive Integrity, Measurement Hardness, Capture Resistance, and Trust Degradation Risk*. Our *goal-by-goal analysis* reveals that **no goal is immune to perverse incentives or manipulation**. For example, narrow quantitative indicators can distort priorities (Goodhart's law) and encourage "gaming" rather than genuine progress ⁵ ⁶ . In some cases, well-meaning targets inadvertently leave the most vulnerable behind ⁷ , or allow powerful actors to appear compliant without meaningful change.
- **Survivability Scores:** We assign each SDG a *survivability score* reflecting its resilience to adversarial conditions. **Most goals score in the low-to-middling range**, indicating substantial risk that hostile actors (or even well-intentioned actors under pressure) could derail, co-opt, or hollow out the goal. As of 2023, only ~15% of SDG targets are on track globally, while 37% are stagnating or reversing – notably on poverty, hunger, and climate action ⁸ . This lack of progress not only endangers outcomes but also **breeds public cynicism** toward the agenda ⁹ .
- **Reconstructed Goals Framework:** To address these issues, we propose **reconstructed versions of all 17 SDGs**, redesigned to meet six stringent constraints: they are **Symmetry-Preserving, Outcome-Bound, Adversarially Auditable, Capture-Minimizing, Failure-Visible, and Trust-Restorative**. These revised goals emphasize equitable outcomes (no "leaving anyone behind"), concrete and verifiable targets, transparency with independent oversight, and mechanisms to detect and correct failures. For instance, *SDG 6 (Clean Water)* is reframed to prioritize universal access verified by independent monitoring, ensuring that improvements reach **all** communities rather than just those easiest to serve.
- **Governance and Safeguards:** We recommend instituting **"kill-switch" criteria** for SDGs – predefined conditions under which a goal or its implementation strategy is reevaluated or retired if it is doing more harm than good (e.g. encouraging bad incentives or showing

persistent lack of credible progress). We also discuss meta-level risks, such as the potential for **institutional capture of the entire SDG apparatus** or erosion of trust in multilateral frameworks, and how our reconstructed goals and oversight processes can mitigate these risks.

In summary, this report finds that while the UN SDGs are well-intentioned, they require significant redesign to withstand adversarial pressures in a fracturing global landscape. The proposed “SDG 2.0” framework preserves the original vision of sustainable development but fortifies it against exploitation, ensuring that progress is real, measurable, and broadly trusted. These recommendations are put forward for scrutiny by sovereign regulators, UN bodies, and independent experts, to initiate a candid dialogue on strengthening global goal-setting against manipulation and failure.

Red Team Methodology

Approach and Threat Model

Red teaming in this context means taking an adversarial perspective to probe how the SDGs could fail or be subverted. Rather than assuming good faith cooperation, we posit a threat model wherein actors – whether national governments, corporations, or other stakeholders – may **intentionally or unintentionally exploit the goals** for self-interest. Key threat vectors include **narrative manipulation, institutional capture, power asymmetries, and hostile incentives**:

- *Narrative Manipulation*: Actors may **misuse the rhetoric of SDGs for image or politics**, touting alignment with goals while obscuring poor performance or harmful practices. For example, an institution might brand its initiatives with SDG logos for prestige while contributing little to real outcomes ¹⁰ ¹¹ .
- *Institutional Capture*: Powerful stakeholders can **capture the processes or indicators** of a goal to serve their interests. This might involve lobbying to dilute targets, cherry-picking metrics that cast them in a favorable light, or dominating partnerships so that investments flow to their priorities. The **co-opting of the SDG agenda** by profit or prestige motives – as observed even in academia and corporate CSR – can undermine genuine progress ¹² ³ .
- *Power Asymmetry*: The SDGs are universal, but actors have unequal power. Wealthier countries or large firms might shift burdens to weaker parties or set global agendas that favor themselves. An adversary could leverage this by **free-riding on others’ efforts** or pressuring for goal interpretations that ignore historical responsibilities (e.g. downplaying developed countries’ duties on climate or inequality).
- *Hostile Incentives*: Perhaps the most insidious threat is **perverse incentives** created by the goals or their indicators. Under pressure to show progress, actors may resort to *gaming metrics* or taking superficial actions. We consider how *Goodhart’s Law* (“when a measure becomes a target, it ceases to be a good measure”) manifests in SDG implementation ⁵ ⁶ . For instance, if a goal is measured by a single quantitative indicator, actors might optimize that number at the expense of the underlying objective – achieving *numeric success but substantive failure*. Such hostile incentive structures can be exploited by anyone seeking an easy win or propaganda victory.

Our analysis assumes adversaries that are **sophisticated and resourceful** – e.g. a government manipulating poverty statistics to claim SDG success, or a corporation publicly supporting “climate action” while quietly lobbying against emissions regulations. We also consider systemic adversarial conditions: global economic shocks, conflicts, or disinformation campaigns that stress-test the SDG

framework's robustness. The overarching question is: *How well do the SDGs survive when not everyone plays fair?*

Five-Axis Evaluation Framework

We developed a five-axis framework to methodically evaluate each SDG under adversarial conditions:

- 1. Symmetry Integrity:** Does the goal uphold **fairness and symmetry** in its formulation and implementation? A goal with high symmetry integrity does not place disproportionate burdens on one group for the benefit of another, and it ensures *"no one is left behind"* in reality, not just in rhetoric. We probe whether each SDG's design might inadvertently reinforce asymmetries or allow the powerful to shift costs to the vulnerable. For example, if progress on an economic goal comes at the expense of environmental degradation in poorer regions, symmetry is violated. Many critics have noted that some SDGs, despite aiming to "leave no one behind," may actually **exacerbate inequalities** or overlook power imbalances ². We evaluate whether adversaries could exploit a goal's design to benefit one population while neglecting or harming another.
- 2. Incentive Integrity:** This axis examines whether the goal's targets and indicators align with **true positive outcomes**, or if they create opportunities for perverse incentives. A goal with high incentive integrity encourages actors to do the *"right thing"* even when gaming is attempted – i.e. the easiest way to score well is genuinely solving the problem. We check each SDG for **perverse incentive risks**: Could an actor meet the letter of the goal while betraying its spirit? Are there shortcuts that look like success but leave underlying issues unresolved? For instance, if a sanitation goal is measured purely by upgrading facilities, governments might focus on improving service for those already served (to hit targets cheaply) while ignoring those with no service at all ⁷. Such behavior yields impressive statistics but widens equity gaps, indicating poor incentive integrity. Our analysis flags such potential distortions.
- 3. Measurement Hardness:** By *hardness* we mean the robustness, clarity, and **auditability of a goal's metrics**. Goals that are **vague or qualitative** (soft metrics) are easier for adversaries to manipulate – progress can be claimed without evidence, and failures can be hidden in ambiguous language. We favor goals with "hard" metrics: clearly defined, quantifiable indicators that can be independently verified. However, even seemingly concrete indicators can mislead if too narrow. We scrutinize if each SDG's progress is meaningfully measurable or if data gaps and indicator over-simplification pose a weakness. Notably, the SDGs rely on 231 global indicators, yet over-reliance on limited quantitative metrics has been warned to *"risk damaging the goals' ambition, integrity, and legitimacy"* ⁵ ⁶. We consider whether adversaries could exploit measurement weaknesses – e.g. by cherry-picking data, exploiting lagging monitoring systems, or adhering to the letter of an indicator that doesn't capture real-world outcomes (a phenomenon described as producing *"fantasy artifacts"* in development metrics ³).
- 4. Capture Resistance:** This axis evaluates how well each goal resists **capture by special interests** or politicization. A capture-resistant goal is formulated and governed in a way that makes it difficult for any single actor or narrow coalition to hijack it for private gain or ideological purposes. We ask: Could a powerful entity **bend this goal to its agenda** without formally violating it? This could happen through dominating the narrative (e.g. framing fossil gas as "clean energy" to satisfy Goal 7), controlling funding flows (tying Goal 17 partnerships to conditional aid or commercial returns), or lobbying to weaken targets. We draw on evidence that certain SDGs have been co-opted in practice – for instance, some universities and corporations use SDG branding for prestige while sidestepping the harder changes needed ¹⁰ ¹³. Each goal

is assessed for features (or lack thereof) that either **mitigate capture** (e.g. multi-stakeholder oversight, conflict-of-interest safeguards) or leave it open to domination.

5. **Trust Degradation Risk:** Finally, we consider the risk that failures or manipulations in a given goal will **degrade public trust** in the broader sustainable development agenda or institutions. Trust is the currency that sustains global cooperation – if people come to see the SDGs as empty words, biased, or consistently unfulfilled, the entire framework loses legitimacy. We evaluate whether an SDG is particularly prone to highly visible failures or controversies that adversaries could use to discredit it. For example, if a goal promises something unrealistic (“end poverty everywhere”) and inevitably falls short, bad-faith actors can point to the failure as proof of incompetence or hypocrisy, fueling apathy or backlash ⁹. There is already evidence of political actors attacking the SDGs as “*misguided, wasteful, or coercive*”, tapping into public frustrations with slow progress ¹⁴. We rate each goal on how a collapse in its credibility (due to unmet promises or scandal) would impact overall trust, and how easily an opponent might trigger or amplify such distrust.

Using this five-axis framework, we performed a **goal-by-goal analysis** of all 17 SDGs. Each goal received an assessment of key failure modes under adversarial conditions and a *survivability score* (low, medium, or high resilience) summarizing how well that goal can withstand adversarial pressures without losing integrity. These scores are not precise measurements but qualitative judgments based on the convergence of the five factors above.

Six Constraints for Reconstructed Goals

Identifying problems is only half of red teaming; the other half is devising solutions or mitigations. For each SDG, we formulated a **reconstructed goal** – essentially, a redesign of the goal’s wording, scope, and governance approach – intended to address the vulnerabilities uncovered. We constrained these reconstructed goals with six design principles to ensure they are robust in the face of the earlier threats:

1. **Symmetry-Preserving:** The new goal formulation must preserve symmetry in obligations and outcomes. This means embedding *equity* into the goal – e.g. explicit focus on the poorest or most marginalized, or proportional responsibilities for those with greater capacity. A symmetry-preserving goal cannot be met by improving averages while leaving a subset of the population behind. It should be impossible to declare success if any major demographic or region is drastically lagging. This guards against exploitation of asymmetries; all actors should see that fairness is built into the target outcome.
2. **Outcome-Bound:** Each reconstructed goal is tied to a clear **outcome** (end-state) rather than vague aspirations or purely process-oriented phrasing. Being outcome-bound means success is defined by achieving a specific, observable condition (e.g. zero people in extreme poverty, X% renewable energy share, etc.) by the deadline. This tight binding to outcomes ensures **accountability** – efforts either result in the outcome or they fall short. It avoids the loophole of claiming success through activities or inputs that may or may not solve the problem. Outcome-bound goals also make it harder to lower the bar; the end conditions are set in concrete terms that an adversary cannot easily redefine.
3. **Adversarially Auditable:** The progress toward the goal must be **auditable in an adversarial context**, meaning that even if some actors prefer to misreport or obscure data, independent verification is feasible. This involves using hard metrics (as discussed) and transparent monitoring systems. Wherever possible, data should be open, **third-party assessments** should be embedded (e.g. civil society or international agencies cross-checking government reports),

and methodologies fixed in advance to prevent indicator-tampering. An adversarially auditable goal is one where cheating or quietly backsliding will likely be exposed. For example, if a goal is to reduce carbon emissions, the reconstructed version would mandate robust emissions accounting and satellite or remote-sensing data to detect discrepancies, rather than relying solely on self-reported figures.

4. **Capture-Minimizing:** The goal's governance and phrasing should minimize avenues for **institutional capture**. This might include structuring the goal as a broad coalition effort with checks and balances, explicitly naming conflict-of-interest safeguards, or framing targets in a way that one industry or faction cannot twist to solely benefit itself. A capture-minimized goal often has **diversified indicators** covering multiple dimensions, so that dominance in one aspect doesn't allow an actor to claim full compliance. It also might involve rotating leadership or oversight by neutral international bodies to prevent nationalistic or corporate hijacking. The idea is to reduce the "surface area" that a malicious actor could latch onto to redirect the goal's intent or resources.
5. **Failure-Visible:** In line with audibility, we require that if progress on a goal is failing, it becomes **highly visible** early on. The reconstructed goals include interim milestones and disaggregated data requirements so that stagnation or regression cannot be easily masked by averages or by delays until 2030. This addresses the common problem of *latent failure* – only discovering at the end date that a goal was far off track. Instead, a failure-visible goal might, for example, trigger alerts if yearly progress falls below a certain rate, or require public reporting on specific lagging sectors. Visibility of failure is crucial for learning and course-correction, and it prevents complacency or false complacency. It also contributes to trust restoration: admitting and seeing problems clearly is the first step to fixing them.
6. **Trust-Restorative:** Finally, each new goal is designed to **restore or enhance trust** in the development framework. This means being honest about trade-offs, avoiding over-promising, and incorporating community voices to build legitimacy. A trust-restorative goal would be communicated with humility and realism – for instance, acknowledging uncertainties – and backed by governance that includes independent watchdogs or beneficiary feedback loops. When people see that a goal is transparently managed and that their concerns are heard (rather than top-down moralizing), trust increases. Also, showing tangible results through the above principles will, over time, validate the process. Essentially, this constraint asks: does the goal, as redesigned, give stakeholders reason to believe in its credibility and in the institutions delivering it? It should actively counter the narrative that "SDGs are just empty UN talk" by demonstrating integrity in practice.

Each of the 17 SDGs has been reconstructed in line with these six constraints. The next sections present our detailed findings per goal, followed by the complete set of reimagined goals and additional cross-cutting recommendations.

Goal-by-Goal Analysis (Vulnerabilities & Survivability)

Below we analyze each original SDG in turn, identifying key adversarial vulnerabilities along the five axes and assigning a **survivability score**. Scores are on a 1–10 scale (10 = highly robust to adversarial interference, 1 = extremely vulnerable). We also summarize why each goal may falter in a hostile scenario. (*The original UN wording of each goal is provided in quotes for reference.*)

Goal 1: No Poverty

Original Formulation: “End poverty in all its forms everywhere.”

- **Symmetry Integrity:** *Moderate.* Goal 1 aspires to eradicate extreme poverty globally, which implies fairness, yet in practice asymmetries emerge. Richer countries and elites may effectively **offload poverty reduction efforts onto poorer nations** or narrow definitions – for instance, focusing only on those just below the \$1.90/day line. If a few populous countries lift large numbers just over the line, global poverty rates fall while deep poverty in marginalized groups persists. This goal can hide internal inequality; an adversary could trumpet national poverty stats while certain regions (or demographics like rural minorities) remain destitute, violating the spirit of “everywhere”. Symmetry is further strained by the fact that poverty is measured by a uniform line that doesn’t account for cost-of-living differences, potentially undervaluing the hardship in high-inequality societies.
- **Incentive Integrity:** *Low.* The primary indicator – the poverty headcount ratio – is highly gameable. Governments face **perverse incentives to game poverty statistics**: they can, for example, alter the poverty line or use outdated population data to show fewer poor. There’s also an incentive to prioritize people closest to the threshold (the “easy wins”) while neglecting the ultra-poor who are harder and costlier to lift. Such tactics would technically reduce poverty percentages, but by **leaving the poorest of the poor behind** ⁷, they undermine the goal’s integrity. The broad wording (“all its forms”) is not matched by equally broad metrics, so important aspects like insecurity or relative poverty can be ignored by focusing narrowly on income.
- **Measurement Hardness:** *Moderate-Low.* Extreme poverty is quantifiable (income < \$X/day), which is a hard measure on the surface. However, the **hardness is undermined by data quality and completeness**. Many low-income countries lack reliable household survey data; if an actor wanted to hide lack of progress, they could delay surveys or dispute methodologies. Moreover, poverty has multidimensional facets (health, shelter, etc.) not captured by income alone – leaving room for disputes about what counts as “ending poverty.” The goal’s sweeping promise (“in all its forms”) is vaguely defined, making it easy for an adversary to claim that certain improvements (like basic income programs) count as success even if overall livelihoods remain precarious.
- **Capture Resistance:** *Moderate.* Because poverty reduction is a universally appealing goal, it is less likely to face overt opposition – but it can be **captured in narrative** by different ideologies (e.g. growth-centric approaches vs. redistribution). A potential capture scenario is institutions promoting particular solutions (say, microfinance or corporate-led initiatives) as *the* SDG 1 strategy, thereby channeling funds and policies to those avenues even if less effective. There is also the risk of **institutional capture via development aid** – wealthy donors could steer poverty programs toward their geopolitical interests rather than where need is greatest. The goal itself is broad enough to accommodate various approaches, which is a strength, but also means without strong coordination, actors can cherry-pick approaches that align with their agendas and still claim SDG alignment.
- **Trust Degradation Risk:** *High.* Goal 1 carries a flagship promise; failure to significantly reduce poverty by 2030 is very visible. If large populations remain in extreme poverty despite years of SDG rhetoric, public trust in global development efforts erodes. Adversaries could highlight cases where poverty persists or has even worsened (especially after shocks like COVID-19) to claim the SDGs are empty promises. The risk is amplified by the absolutist wording “end poverty everywhere” – an impossible ideal that, if unmet, can breed cynicism ⁹. Communities that see

little change in their plight may lose faith in institutions, a sentiment hostile actors can exploit to fuel anti-establishment or anti-aid narratives.

Survivability Score: 4/10. *Goal 1 is vulnerable to data manipulation and selective achievement. While morally unassailable, its broad scope makes it easy to claim partial victory without structural change. Under adversarial conditions – where actors seek cheap wins – there is a real risk of hollow success (statistical poverty reduction without true emancipation), which could be exposed and damage the agenda’s credibility.*

Goal 2: Zero Hunger

Original Formulation: “End hunger, achieve food security and improved nutrition, and promote sustainable agriculture.”

- **Symmetry Integrity:** *Moderate-Low.* Hunger is predominantly an issue in poorer regions, yet the goal’s success depends on global agriculture and trade systems often controlled by richer actors. This asymmetry means **food security improvements can be uneven** – e.g. increasing exports from industrial farms might lower global hunger stats but bankrupt small farmers, or hunger might decrease in stable regions while conflict zones see famine. Adversaries (or even well-meaning governments) might allocate food resources to politically important areas while neglecting marginalized groups – achieving national food security on average but leaving pockets of severe hunger (thus breaking symmetry). Also, heavy focus on staple crop yields could ignore nutritional quality for vulnerable groups (like children needing micronutrients), meaning “calories per capita” rises while malnutrition persists.
- **Incentive Integrity:** *Low.* The emphasis on sustainable agriculture and productivity carries potential **perverse incentives**. Governments could chase higher crop output at all costs – e.g. expanding monocultures or using intensive inputs – to show progress on hunger, inadvertently incentivizing unsustainable practices that undermine long-term food security (soil depletion, water overuse). There’s also incentive to focus on quantity over access: a country might produce record harvests (improving food availability metrics) yet poor households still cannot afford food. An adversarial actor might hoard or divert food aid for profit or political leverage, while still meeting export or production targets. Current indicators (prevalence of undernourishment, stunting rates, etc.) are lagging and can be gamed through short-term measures (like import surges) that don’t last, just to hit targets.
- **Measurement Hardness:** *Moderate.* Hunger and nutrition have measurable facets (caloric intake, child growth metrics). However, **data is often sparse and averaged**. An authoritarian regime could suppress famine data or refuse surveys, masking severe hunger. The composite nature of “food security” is hard to capture; one indicator can improve while another worsens. For example, undernourishment rates might drop (if fewer calories short) but acute malnutrition could rise in war-torn pockets – which might be obscured in national averages. Nutrition quality (“hidden hunger” of micronutrient deficiency) is particularly hard to measure at scale, giving adversaries wiggle room to declare victory (“hunger is solved” in calorie terms) while significant nutrition problems persist. The inclusion of “sustainable agriculture” adds complexity: measuring sustainability (like soil health, biodiversity on farms) is not straightforward, so this part may be ignored in favor of easier hunger metrics.
- **Capture Resistance:** *Low.* The global food and agriculture arena is **prone to capture by powerful agribusiness interests**. There is a risk that big agri-corporations and large agrarian countries steer Goal 2 initiatives toward high-tech, high-input farming (which benefits their products and trade interests) rather than agro-ecological or local solutions. Corporate capture

could mean the SDG narrative pushes GMO seeds, synthetic fertilizers, and global value chains as the solution to hunger, sidelining indigenous practices or food sovereignty approaches. Similarly, food-exporting nations might capture the agenda to open markets, claiming it's about food security. The "sustainable agriculture" aspect could be co-opted as well – e.g. large players marketing slightly more efficient industrial farming as 'sustainable' while crowding out truly regenerative methods. Without careful governance, Goal 2 can become a banner for **business-as-usual with green rhetoric**, rather than genuinely transforming food systems.

- **Trust Degradation Risk:** *High (in crisis scenarios).* Food is visceral; if Goal 2 is seen to fail (e.g., images of starvation while global promises were made), it sharply erodes public trust. Conversely, declaring success amid hunger can appear deeply out-of-touch or propagandistic, fueling anger. Adversaries can exploit any food crisis – say a spike in prices or famine – to claim the SDGs are irrelevant. For instance, the fallout of global events (pandemics, wars) causing hunger in 2025-2030 could severely undercut belief in the "zero hunger" ideal. Also, if policies under the SDG banner lead to unintended harm (like land grabs for large farms or higher food prices due to export-oriented strategies), affected communities will distrust the goal. *Hunger* has moral weight; visible hypocrisy or shortfall here can taint the credibility of the broader UN development effort.

Survivability Score: 3/10. *Goal 2 is highly vulnerable: powerful interests can hijack agricultural agendas, and numeric gains are achievable without truly ending hunger for all. In adversarial conditions, one could meet production targets yet still "weaponize" food (through hoarding or selective distribution). The goal's broad scope (from hunger to sustainable farming) is a strength aspirationally but a weakness in accountability. Without major safeguards, Goal 2's noble aim could be undermined and become a lightning rod for criticism when hunger inevitably persists in some quarters.*

Goal 3: Good Health and Well-Being

Original Formulation: "Ensure healthy lives and promote well-being for all at all ages."

- **Symmetry Integrity:** *Moderate.* Health inequities are rampant – across countries and within them. Goal 3 strives for "health for all", but adversaries (or even normal politics) can lead to **skewed health outcomes**. Resources might flow to high-profile issues (like hospitals in major cities) while rural or minority populations lack basic care. A country could improve average health indicators (e.g. life expectancy) even as gaps widen between rich and poor. Symmetry is especially tested in global health crises: wealthy nations could monopolize treatments (as seen with vaccine access disparities), undermining the universality of healthy lives for all ages. If powerful stakeholders prioritize diseases that affect them (say, lifestyle diseases) and neglect those of the poor (tropical diseases), the goal's integrity is compromised. Adversarial manipulation could involve cherry-picking whose health improves – an authoritarian might lavish care on military and urban elites for productivity, ignoring marginalized communities, yet still claim national health progress.
- **Incentive Integrity:** *Moderate.* Many health targets (reducing mortality, increasing immunization, etc.) are positive-sum if pursued sincerely. However, **perverse incentives** lurk in how outcomes are achieved. Healthcare systems might chase numeric targets (like number of vaccinations or hospital beds) rather than quality of care. For example, if maternal mortality is a key indicator, a government might invest in better reporting or in emergency obstetric care in hospitals (helpful) but neglect broader women's health and prenatal care that are harder to measure – or worse, discourage reporting of maternal deaths to make stats look better. There's also incentive for "vertical" health programs (targeted at specific diseases with external funding)

that tick SDG boxes, while systemic issues (like healthcare workforce or primary care access) languish because they don't have singular metrics. Pharmaceutical companies and donors can influence priorities: an adversarial interplay here could be pushing expensive interventions that improve certain indicators (like a new drug that reduces one disease metric) while cheaper, broader public health measures (sanitation, prevention) are underfunded.

- **Measurement Hardness:** *Moderate.* Health data is among the better-tracked (births, deaths, disease incidence), but **measurement issues remain**. Some indicators are modeled with uncertainty (especially in low-income settings for mortality rates). An actor could manipulate data – e.g. under-reporting disease outbreaks or deaths (as has happened with some pandemics) – to maintain an illusion of progress. Well-being is inherently vague: beyond health, “well-being” could be interpreted variously; this softness allows a government to claim improvements via, say, a happiness survey or mental health program without much accountability. The breadth of Goal 3 (covering maternal health, epidemics, substance abuse, etc.) means progress can be uneven; one can cherry-pick successes in easier areas while ignoring harder ones. For instance, it's easier to mass-distribute bed nets for malaria (measurable outputs) than to improve mental health or reduce non-communicable diseases tied to lifestyle – an adversary could tout the former and quietly sidestep the latter.
- **Capture Resistance:** *Moderate-Low.* The health sector is subject to capture by both **political and commercial interests**. Politically, leaders love cutting ribbons for new hospitals or launching disease campaigns – they may skew health spending towards visible projects that garner support, rather than what is truly needed (primary care in villages might be less visible). Commercially, pharmaceutical and private healthcare lobbies can shape which health issues get attention and funding. For example, SDG 3 could be captured to push expensive medications or technologies while basic public health (not profitable to companies) is neglected. There is also a risk of international capture: donor nations or foundations often drive health agendas in developing countries (consider how HIV, TB, malaria got specific focus in MDGs). If their interests diverge from local needs, health priorities under SDG 3 might be misaligned. In an adversarial scenario, one could imagine a company influencing health guidelines to favor its product (for “well-being”) or an autocrat emphasizing health programs that boost their image (like elite hospitals) at the expense of broad care.
- **Trust Degradation Risk:** *Moderate.* Health is personal; failures or scandals here can rapidly erode trust. If, for instance, a vaccine campaign goes wrong or a promised improvement (like universal health coverage) doesn't materialize, people become distrustful not just of healthcare but of international promises. The SDG aim of health for all could be mocked by opponents whenever there's a glaring health crisis (e.g., “They said well-being for all ages, yet look at the COVID-19 devastation or opioid epidemic!”). On the other hand, health successes are also tangible and can build trust. The risk lies in overpromising (saying we will end epidemics by 2030 – unlikely for, say, AIDS or tuberculosis) and then failing, which gives ammunition to critics. Also, heavy-handed health mandates (if perceived as tied to global agendas) can feed conspiracy narratives, as seen by populist pushback during the pandemic against global health guidelines¹⁴. So, missteps in Goal 3's implementation could degrade trust in both health authorities and the UN framework.

Survivability Score: 5/10. *Goal 3 has moderate resilience: the health domain benefits from strong data systems and widespread buy-in, yet it is not immune to manipulation. Under adversarial strain, one might see superficial health gains (like improved reported stats) that mask inequity or neglect. Still, compared to some goals, health goals often have clearer technical paths and multi-actor support, so outright subversion is*

somewhat harder. The score reflects a mix of solid progress in areas like child survival (raising credibility) and serious concerns in areas like global pandemic response (where failures can dramatically set back trust).

Goal 4: Quality Education

Original Formulation: “Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all.”

- **Symmetry Integrity:** *Moderate-Low.* Education access has expanded, but **inequities persist** between urban and rural, rich and poor, boys and girls. An adversary could meet the letter of inclusion by enrolling students in primary school (boosting enrollment ratios) while quality and completion gaps remain – e.g. marginalized children might be the first to drop out or receive substandard schooling. Thus, average schooling years can rise while the poorest or minority groups still lag far behind in learning, compromising symmetry. Additionally, if a country emphasizes elite universities or certain regions’ schools to showcase success, it may widen the gap with underserved areas. Equitable education is hard to measure beyond enrollment – e.g. are disabled children or linguistic minorities truly included? Without specific safeguards, these groups can be neglected. In a power asymmetry sense, wealthy families can always supplement education (private tutoring, etc.), so if public education quality isn’t uniformly improved, SDG 4’s broad promise doesn’t hold symmetrically.
- **Incentive Integrity:** *Low.* The dominant indicators for education are **years of schooling and enrollment rates**, which create a classic **perverse incentive**: focus on quantity over quality. Governments have an incentive to get kids into classrooms (often a relatively straightforward policy task) but not necessarily to ensure they learn. This has led to what experts call a “learning crisis” – many children attend school but learn shockingly little. For example, even before COVID-19, **over 50% of children in low- and middle-income countries could not read a simple text by age 10** ¹⁵ ¹⁶, despite schooling – a stark indicator that schooling targets were met at the cost of actual learning outcomes. Adversarially, a government could reduce standards or inflate graduation rates to claim SDG progress (“more graduates this year!”) even if those graduates lack basic skills. There’s also incentive to emphasize easy-to-teach curricula geared to tests rather than critical thinking, reducing education to test prep to hit “quality” indicators like literacy/numeracy tests. Such gaming produces paper results but hollow educational improvement.
- **Measurement Hardness:** *Moderate.* Enrollment and completion are hard numbers, but **quality of education is difficult to measure** succinctly. International assessments (like PISA or national exams) give some gauge of learning, but not all countries participate, and results can be massaged (e.g., excluding certain students from tests). Lifelong learning opportunities are even vaguer – how to measure “opportunities for all” in adult education? This softness allows an actor to claim various initiatives as fulfilling SDG 4 without clear benchmarks. Data on skills and learning is often delayed and sample-based. A savvy adversary could tout inputs (built 100 new schools, hired teachers) and avoid discussing outputs (learning outcomes). The reliance on self-reported data for things like literacy can be manipulated (e.g., defining literacy loosely). Therefore, while we have lots of education data, it may not capture the true state of learning, providing cover for complacency or over-optimistic assessments.
- **Capture Resistance:** *Moderate.* Education, being largely managed by public sector in most countries, is somewhat less prone to corporate capture, but it can be **ideologically or politically captured**. A regime can shape curricula to its narrative (e.g., omit inconvenient history, promote propaganda) all while claiming to advance “quality education.” There’s also increasing private

sector influence – edu-tech companies and private school chains could push solutions that align with their profit motive, capturing the discourse on “innovative learning” or “digital learning for all” as panaceas, possibly at the expense of proven basics. If international donors emphasize certain educational models (say STEM or digital skills) because of global trends, local priorities like basic literacy or indigenous knowledge might be sidelined. In adversarial terms, one could imagine a scenario where an entity promotes a narrow curriculum that serves its interests (for example, training for specific labor needs) under the guise of SDG 4, hence capturing the outcome to fill its talent pipeline while broader educational goals suffer.

- **Trust Degradation Risk: Moderate.** Education is something that communities deeply value; visible failures (like many graduates being unemployed or functionally illiterate despite schooling) can spark public outcry. If SDG 4 efforts lead to lots of shiny new schools but parents see that children still aren’t learning useful skills, trust in both the education system and the global agenda erodes. However, education improvements are generally slow and incremental – the risk is a creeping disillusionment rather than a sudden crisis. That said, youth unemployment or mismatch of skills can become a political flashpoint (“We got educated under SDG push but have no jobs or practical skills”). An adversary could exploit this by claiming the education goal wasted a generation’s time on irrelevant schooling. On a larger scale, if after 15 years the world is still far from “quality education for all,” it questions either the commitment or the feasibility of such global goals, feeding cynicism. Trust in institutions declines when people feel promises of opportunity (which education symbolizes) were broken.

Survivability Score: 4/10. *Despite global enthusiasm for education, Goal 4 has notable weaknesses under adversarial conditions. It is easy to fulfill on paper (build schools, enroll students) but difficult to ensure genuine learning and equity. This gap provides adversaries ample room to claim victory or subvert content without detection. The score reflects the concern that without stricter focus on learning outcomes and equity, SDG 4 could be superficially achieved yet substantively failed, undermining confidence in educational initiatives.*

Goal 5: Gender Equality

Original Formulation: “Achieve gender equality and empower all women and girls.”

- **Symmetry Integrity: Moderate.** Gender equality by definition is about symmetry (equal rights, opportunities, burdens between genders). The goal calls out women and girls as the disadvantaged group globally, aiming to elevate them to parity. However, adversaries can selectively interpret equality. For instance, a country might promote women’s representation in politics or business (benefiting relatively elite women) while neglecting poor women or issues like unpaid care work – creating an illusion of symmetry achieved at the top echelons, while vast asymmetry remains in daily life. Also, focusing exclusively on women could cause some to claim “reverse asymmetry” if men in certain areas (e.g., education dropouts or certain health issues) are struggling – an adversary might exploit that narrative to undermine the goal’s legitimacy. True symmetry requires bringing all genders to a balance, but an opponent could paint Goal 5 as one-sided (“only cares about women, not men”), sowing division. Additionally, intersectional factors (race, class) may mean some women (urban, rich) gain equality while others (rural, minority) do not – symmetry within the gender category is at risk if policies favor those easier to empower.
- **Incentive Integrity: Moderate-Low.** Governments have found that *formal* gender equality (laws, quotas) is relatively easier than *substantive* equality. Thus, a **perverse incentive** exists to do the minimum: pass laws, set quotas, launch awareness campaigns – all ticking the SDG 5 boxes –

without enforcing real change or providing resources for implementation. For example, making gender discrimination illegal (a target) can be done by decree, but ensuring it's enforced in every village is hard. Under adversarial conditions, a state might showcase some high-profile female appointments or accolades to signal progress while suppressing women's rights activists or underfunding enforcement of protections. There's also a risk of tokenism: promoting a few women to visible roles (the incentive being global praise) rather than systemic change like education and economic empowerment at scale. In some cases, regimes could co-opt women's empowerment language to sanitize their image ("look, we have women driving now!") while jailing feminist critics – achieving SDG indicators but undermining genuine empowerment (the incentive being reputation rather than actual equality).

- **Measurement Hardness:** *Moderate.* Some aspects of gender equality are measurable (e.g., female labor force participation, wage gap, seats in parliament held by women). These provide hard numbers that can track progress. However, **many crucial aspects are qualitative or hard to capture:** gender-based violence, social norms, intra-household power, sexual and reproductive health rights in practice, etc. An actor can tout improvements in easy-to-measure areas (say, number of women in leadership) and ignore things like domestic violence rates or maternal mortality disparities. Data on sensitive issues (like violence or unpaid work) often relies on surveys that can be influenced by stigma or underreporting. Hence, a government could claim ignorance ("we have no data on that, so no problem") or manipulate the narrative by only highlighting favorable metrics. The broad empowerment goal also lacks a single headline metric, making it easier to cherry-pick. This softness benefits an adversary who wants to claim "mission accomplished" by selectively measuring progress, while the lived reality of most women might tell a different story.
- **Capture Resistance:** *Moderate.* Gender equality efforts can be politically contentious; however, direct *capture* by corporate interests is less obvious here (except maybe companies promoting certain "women's empowerment" programs for PR). More likely is **cultural or political capture:** the goal can become a pawn in ideological battles. For example, in some settings, conservative forces might "capture" the interpretation by limiting it to what they deem acceptable (e.g., promoting women in traditional roles as mothers/educators as their form of empowerment, while rejecting aspects like reproductive rights or LGBTQ+ inclusion). Internationally, there's a tension – some autocratic regimes publicly support women in STEM or government as a modernizing image, even as they constrain women's personal freedoms; they capture the narrative to suit regime legitimacy. Alternatively, there's a risk of capture by a narrow Western narrative of feminism that doesn't resonate locally, which opponents then use to claim SDG 5 is "cultural imperialism." In short, Goal 5 can be co-opted either to promote a regime's image or attacked as alien, depending on who seizes the narrative.
- **Trust Degradation Risk:** *Moderate.* The fight for gender equality often faces backlash. If SDG 5 is seen as unsuccessful – e.g., if everyday women feel no improvement or even perceive situations worsening (like more overt violence or pushback) – it can cause disenchantment among the very people it's meant to empower. On the other hand, patriarchal or populist elements might stoke distrust by framing SDG 5 as a threat to cultural values or men's rights. We already see some politicization of gender issues; a concerted adversarial effort could portray global gender initiatives as nefarious (for instance, extremists painting women's empowerment as part of a "Western agenda"). Failure to deliver tangible changes (like economic opportunities or safety) will certainly reduce trust in institutions championing gender equality, especially among women who risked social capital for promised support. Conversely, any success builds trust – so the risk is if the goal is co-opted or stagnates, it harms credibility of both governments and international advocates in the eyes of the public.

Survivability Score: 5/10. *Goal 5 stands on contentious social terrain. It has moderate survivability: legal and institutional reforms can be monitored (so blatant backsliding is visible), and there is strong global normative support for women's rights making outright sabotage costly. Yet, subtle undermining and tokenism are prevalent threats. The score reflects a balance – real progress is possible with alignment of incentives, but cultural resistance and superficial compliance can significantly thwart true equality if adversarial forces prevail.*

Goal 6: Clean Water and Sanitation

Original Formulation: “Ensure availability and sustainable management of water and sanitation for all.”

- **Symmetry Integrity:** *Moderate-High.* Water and sanitation improvements typically target the underserved, often rural poor or urban slums, aiming for universal access. The principle of “for all” directly pushes symmetry. However, adversaries could undermine this by focusing on easier-to-serve populations – e.g., expanding coverage in well-off urban areas (where infrastructure is simpler to build and politically rewarding) while remote communities remain without services. This creates an asymmetric outcome: overall coverage rises, but the gap between best and worst served widens. Another asymmetry risk is between water quantity vs. quality – a government might ensure everyone *nominally* has a water source, but perhaps poorer areas get substandard or intermittent supply. Similarly, sanitation might improve via basic latrines in many places, but only wealthier areas get safely managed sanitation (sewerage, treatment). If those disparities persist, the goal's equity ethos is undercut. **Symmetry is strongest when the last mile (the hardest-to-reach people) are equally prioritized**, which adversarial logic might resist due to higher cost/less visibility of those investments.
- **Incentive Integrity:** *Moderate-Low.* The structure of water/sanitation targets can encourage certain shortcuts. For instance, if progress is measured by proportion of population with “safely managed water” and “safely managed sanitation,” governments may find it most cost-effective to upgrade existing basic services to higher levels, rather than extending services to those with nothing ⁷. This is a known perverse incentive: it's easier to move someone from an intermediate level to safely managed than to provide first-time access to a remote village. Thus, an adversary might do the low-hanging fruit upgrades to boost the stats quickly, leaving the truly unserved behind (contrary to the goal's intent). There's also an incentive to favor large infrastructure (dams, treatment plants) that yield big numbers (millions served) over community-scale solutions that might be more sustainable but yield smaller immediate stats. Additionally, maintenance can be neglected – an adversary might install thousands of water pumps or toilets to count coverage, but if half break down in a few years due to no maintenance, the initial stats were misleadingly positive. Current global monitoring might not catch that quickly, so the incentive is to build (visible) rather than maintain (less visible).
- **Measurement Hardness:** *Moderate.* Access to water and sanitation is measured by tiered indicators (e.g., safely managed, basic, limited, unimproved). These are fairly specific definitions, so progress can be quantified. However, **data collection is a challenge** – it relies on surveys and administrative data. A government inclined to cheat could over-report the number of villages with wells or households with toilets, especially if verification is weak. Also, “sustainable management” of water is broad: measuring water quality, ecosystem health of water sources, and efficiency of use introduces softer metrics. It's easier to measure infrastructure presence than actual service quality (e.g., continuous water supply or clean drinking water free of contamination). Thus a country might claim high access rates while water is frequently rationed or contaminated – a gap adversaries could exploit to appear compliant. Sanitation data similarly

might not capture usage (toilets built but not used, or sewage not properly treated). Without rigorous audits, these gaps allow room for manipulation. On the plus side, failures (like a cholera outbreak due to poor sanitation) are hard to hide entirely, but the lead-up indicators might not flag issues early.

- **Capture Resistance:** *Moderate.* Water is often a public good managed by governments or community providers, which limits direct private capture, but there is still risk of **institutional capture through privatization or monopolies**. For example, a private water company could dominate urban water provision and have a vested interest in SDG projects – it might push for centralized solutions that benefit its business, possibly sidelining cheaper community solutions. In some places, local elites capture water infrastructure (ensuring their area has continuous supply while others ration). Politically, capture can occur if officials treat water projects as patronage (allocating them to allies or popular regions to gain support, not purely by need). Sanitation often suffers from low political priority (it's less glamorous), so an adversary could capture budgets by diverting funds to more visible sectors, leaving sanitation under-resourced. Large infrastructure projects (dams, treatment plants) could be lobbied for by construction interests under the banner of SDG 6, even if smaller distributed solutions might serve the poor better. So while not as capture-prone as some goals, there are avenues where special interests (private utilities, construction firms, local politicians) can skew implementation to their advantage.
- **Trust Degradation Risk:** *Moderate.* Water and sanitation improvements are tangible and directly improve lives, so successes here can greatly boost trust in development efforts. Conversely, **failures are also very tangible** – if promised clean water doesn't materialize, or if newly built toilets are non-functional, communities quickly become disillusioned. Broken water pumps and unfinished sewer projects can become symbols of broken promises. Adversaries could amplify these failures ("look, the UN/ government said everyone would have water, but our village still doesn't!") to undermine confidence. Additionally, if water initiatives lead to unintended harm – e.g., large dam projects displacing communities, or water tariffs rising after privatization – people might feel betrayed by SDG 6 efforts, perceiving them as serving others' interests. However, because water/sanitation is less politicized ideologically than, say, climate or gender, the trust risk is more about delivery failure than active opposition. Still, at a basic level, not having safe water is such a visceral grievance that continued lack thereof in 2030 would make many question the value of global goals.

Survivability Score: 6/10. *Goal 6 is one of the more straightforward and engineering-focused goals, which gives it some resilience. It's harder to completely fake a water connection – either the tap flows or it doesn't – so blatant lies are limited. Many countries have made steady progress here, showing the goal's feasibility with will and funding. However, our adversarial lens spots risk of inequitable implementation and short-termism (build-but-don't-maintain syndrome). Overall, with moderate vigilance, SDG 6 can survive much of the adversarial gaming, hence a slightly above-average score, but it still needs better incentives to truly reach the last-mile communities and sustain services long-term.*

Goal 7: Affordable and Clean Energy

Original Formulation: "Ensure access to affordable, reliable, sustainable and modern energy for all."

- **Symmetry Integrity:** *Moderate.* Energy access has a clear equity dimension – hundreds of millions lack electricity, mostly in poorer regions, and many rely on dirty fuels for cooking. The goal seeks universal access, which inherently addresses a symmetry gap. However, in practice, asymmetries can arise: an adversary might prioritize extending electricity grids to easy-to-reach

or economically viable areas (cities, industries) while remote, impoverished communities still languish off-grid. So one could achieve high national electrification (benefiting most, boosting averages) yet leave behind the hardest cases (the last few percent, often the poorest). Also, the “affordable” aspect is vital – an area could be connected but if prices are high, the poor still can’t benefit. If energy infrastructure is built but usage costs are not subsidized for the poor, asymmetry persists. Another risk: a country could meet its own energy needs by exploiting another’s resources (e.g., building upstream hydropower affecting downstream communities’ water, or land-intensive biofuel plantations affecting local livelihoods) – creating cross-border or rural-urban asymmetries where one group’s clean energy gain is another’s loss. Ensuring symmetrical benefits (everyone gets access and no one bears unfair externality costs) is challenging and can be manipulated by those with power.

- **Incentive Integrity:** *Low-Moderate*. **Perverse incentives** exist around the definitions of “clean” and “modern” energy. For instance, a government might meet renewable energy targets by focusing on megaprojects (like a big hydro dam or commercial solar farm) which count toward sustainable energy statistics but may not actually provide affordable access to the poorest (power might go to industry or export). There is also an incentive to relabel certain energy sources as “clean” to get credit – e.g. pushing *natural gas* or even “clean coal” as affordable, modern energy, which is arguably against the spirit of sustainability but could be justified as improving over fuelwood or diesel. Adversarial actors, especially fossil fuel interests, may lobby for looser definitions so that their products count (we’ve seen efforts to include gas or carbon-capture coal in sustainable energy mix). If metrics only look at percentage of renewable energy, countries might neglect the affordability part – e.g. renewables grow, but energy prices spike, hurting the poor (but the indicator looks good). Also, focusing on electricity could ignore clean cooking – governments might electrify villages (to hit “access” goals) but not help households transition away from firewood for cooking, which is a hidden failure if not measured. The incentive might be to do the easier thing (grid connections) and claim victory while the harder problem (clean cooking solutions) lags.
- **Measurement Hardness:** *Moderate*. We can measure electricity access rate, renewable share in energy, and energy intensity. Those are relatively concrete. However, **affordability and reliability are harder to quantify** uniformly. A country could report 100% access, but if supply is intermittent or outages common, the goal isn’t truly met – yet the primary metric might miss that. Affordability might be gauged by percent of income spent on energy, but data on that is patchy; an adversary could downplay price issues. Also, in measuring “clean,” there’s some complexity: should large hydro count (it’s renewable but can have social/environmental downsides), do improved cookstoves count as clean cooking? The way metrics are set could be gamed (e.g., calling something clean when it’s borderline). Another subtlety: emissions reduction might be measured under climate goals, but not directly under Goal 7, so a country could meet Goal 7 by giving everyone energy (which is good) but via cheap coal power (violating sustainability in a broader sense) – if sustainability isn’t robustly defined, that’s a loophole. Thus, while access numbers are “hard,” the qualitative aspects of the goal open measurement loopholes.
- **Capture Resistance:** *Low-Moderate*. The energy sector is **highly susceptible to capture by powerful interests** – fossil fuel companies, utility monopolies, even state-owned enterprises with political clout. There is a risk that these actors influence the agenda such that certain solutions (ones that benefit them) are prioritized. For example, a major oil/gas company may push natural gas expansion as a way to achieve “modern energy for all,” capturing policymaking so that gas infrastructure is funded under SDG 7 (with the argument it’s needed for affordable access), at the expense of decentralized renewables that might empower communities more but

offer less profit to big players. Renewables themselves can be captured – large solar/wind projects often involve big firms and land deals; without safeguards, local communities might not have a say. Additionally, if energy is politicized (as in some countries where electricity is given for free or subsidies are manipulated for votes), the goal can be captured as a populist tool, giving short-term handouts but not sustainable systems. Internationally, financing for Goal 7 could be captured by donors promoting their own industries (e.g., a country funding solar panel projects but only using panels from its domestic manufacturers). All these reduce the goal's impartial pursuit.

- **Trust Degradation Risk: Moderate.** Energy, like water, is something people notice directly. If SDG 7 efforts succeed – lights turn on, fuel is cleaner – it builds trust. But failures or perceived betrayals (like price hikes or blackouts) can quickly cause public anger. For instance, if a government touts “electricity for all” but communities experience constant outages or huge bills, citizens become cynical about the promises. Also, the “sustainable” part if ignored can backfire on trust in the long run – e.g., communities might initially welcome a coal plant for jobs and power, but later suffer pollution; they would feel misled about clean energy claims. On the global stage, if the world continues toward climate breakdown (Goal 13 failure) while claiming progress on clean energy, youths and activists may view SDG 7 as greenwashing, hurting trust in international climate and development institutions. Conversely, if pursuit of SDG 7 leads to unpopular measures (like removing fuel subsidies for sustainability), that can trigger public backlash (as seen in protests over fuel price reforms), which adversaries might exploit to say “these global goals hurt you.” Therefore, mismanaging the balance of affordable vs. sustainable could degrade trust in the goal.

Survivability Score: 5/10. *Goal 7 is a mix of technical feasibility and political economy risk. Technologically, delivering energy for all is achievable; the main issues are about how and for whom. Under adversarial conditions, one might see progress in access but skewed to certain groups, or “clean” energy being only somewhat clean due to lobbying. The sector's powerful incumbents and the tension between short-term affordability and long-term sustainability put this goal at moderate risk. It's neither extremely fragile nor robust without active safeguards – hence a middling score.*

Goal 8: Decent Work and Economic Growth

Original Formulation: “Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all.”

- **Symmetry Integrity: Low-Moderate.** Economic growth often comes with **winners and losers**, and Goal 8's broad scope makes symmetry a challenge. While it calls for *inclusive* growth and decent work for all, adversarial realities mean growth can be very uneven – e.g., growth concentrated in certain sectors or regions (tech hubs, cities) with wealth not trickling down. A country could hit high GDP growth (satisfying the goal superficially) yet inequality soars and many remain in precarious work, violating the “for all” intent. If powerful actors drive growth (corporations, capital owners), they may capture most gains, leaving labor with stagnant wages – effectively breaking symmetry between profits and workers' welfare. Additionally, a focus on aggregate growth might overshadow the need for equity; an adversary could even justify policies that hurt some groups (like deregulating labor protection “for growth”) which boosts GDP but harms workers' rights. The symmetry between present and future generations (sustainable growth) can also be ignored – burning through natural capital for current growth, leaving future generations poorer (an asymmetry across time). So without strong safeguards, this goal can mask significant asymmetries under the sheen of growth.

- **Incentive Integrity:** *Low*. The core indicator for economic growth is GDP, which is infamous for not distinguishing quality or distribution of growth. **Goodhart's law looms large:** if GDP is targeted, governments have incentive to maximize it at almost any cost. That could mean encouraging industries with high output but negative social/environmental externalities (polluting factories ramp up GDP, for example), or even statistical gaming (revising how GDP is calculated to inflate it). There is also a temptation to prioritize quantity of jobs over quality: pushing people into any employment (even if informal, low-paid) to reduce unemployment stats while ignoring whether it's "decent work." An adversarial approach might promote sweatshops or gig economy jobs that are exploitative yet claim success on employment numbers. The goal's phrasing of "sustained growth" might incentivize unsustainable borrowing or resource extraction to keep growth figures up short-term. Moreover, "full employment" could be pursued via expanding low-paid public works or manipulating definitions of employment. The multiple aims (growth, inclusion, sustainability, employment, decent work) create trade-offs that a bad-faith actor can choose from – likely focusing on the easiest to achieve (growth and basic employment) and neglecting the harder (inclusion, sustainability, decency). Indeed, critics note a potential **contradiction between perpetual GDP growth and sustainability** ¹⁷; adversaries will likely resolve that by favoring growth metrics and sidelining sustainability if it's inconvenient.
- **Measurement Hardness:** *Moderate*. GDP growth is a hard number (though subject to revision and some manipulation). Employment rates are also measurable (with caveats like underemployment or informal work). "Decent work" and "inclusive growth" are much softer – they involve quality of jobs, fairness of income distribution, safety, etc. There are proxy indicators (like median income, working poverty rate, labor productivity, etc.), but a government can cherry-pick which ones to highlight. If GDP is doing well but, say, the share of income to the bottom 40% is shrinking (a sign growth isn't inclusive), an adversary might bury that statistic. Many developing countries struggle to measure informal economy and labor conditions, giving leeway to avoid uncomfortable truths. Also, sustainability of growth isn't easily captured by short-term indicators – a country could be depleting forests or degrading labour rights to fuel current growth, which won't show up in GDP until later (and not directly unless a crisis occurs). That delayed effect means the metrics can look fine while a bubble builds or social tension rises. So while some parts of Goal 8 are measured rigorously, the holistic picture is not, enabling selective storytelling.
- **Capture Resistance:** *Low*. Economic policy is often shaped by powerful financial and corporate interests. **Capture risk is high:** large corporations and investors typically influence growth agendas, potentially at odds with "decent work for all." They might lobby for deregulation, tax breaks, and labor flexibility that boost growth but undermine worker rights or widen inequality. Under SDG rhetoric, a government might form "public-private partnerships for growth" that end up privileging big businesses – effectively capturing the development narrative to serve the business-as-usual growth model. Internationally, institutions like the World Bank or IMF might push policy reforms (privatization, free trade) under the guise of promoting SDG 8 growth, aligning with certain economic ideologies that benefit capital owners. Meanwhile, labor unions or workers' voices (crucial for decent work) might be sidelined. In adversarial hands, Goal 8 could become a cover for **neoliberal reforms** that arguably conflict with inclusion/sustainability aspects ¹⁸ ¹⁹. Also, unsustainable industries (oil, mining) might capture the "growth" narrative to justify continued expansion despite environmental harm, claiming they are creating jobs and GDP (short-term gains capturing policy at long-term cost).
- **Trust Degradation Risk:** *Moderate-High*. People's trust in institutions strongly correlates with economic well-being. If SDG 8 initiatives lead to perceived injustices – say, growth that benefits only elites or policies that cause layoffs/price spikes in the name of reform – public backlash can

be severe. Many political upheavals stem from grievances about unemployment or inequality, which this goal directly touches. For instance, if governments tout SDG progress because GDP is rising, but ordinary people see stagnant wages and precarious jobs, it breeds cynicism and anger. The goal's broad promises could seem hollow, fueling populist narratives that international agendas only serve the global elite. Indeed, stagnation or decline in decent jobs while leaders celebrate “growth” is a recipe for eroded trust. Additionally, pushing indefinite growth in a world of climate concerns can prompt younger generations to distrust the SDG framework as being out-of-touch or complicit in environmental destruction (a moral hypocrisy issue). There is already academic and activist critique that **pursuing high growth is incompatible with ecological limits**, and if SDG 8 is seen as overriding climate commitments, it could damage trust in the coherence of the UN's agenda ¹⁷. Thus, mismanagement or capture of Goal 8 can undermine faith in both economic governance and sustainable development as a whole.

Survivability Score: 3/10. *Goal 8 is one of the most contentious under adversarial analysis. It sits at the intersection of economic powers and social outcomes, making it highly vulnerable to being co-opted by growth-for-growth's-sake narratives. Without significant refocusing (e.g., prioritizing decent work and equitable distribution over raw GDP), the goal can be achieved in a Pyrrhic way (numbers up, people left behind). Its internal conflicts (growth vs sustainability vs equity) are easily exploited. As it stands, under hostile incentives, Goal 8 could both fail its true intent and erode trust, giving it a low survivability score.*

Goal 9: Industry, Innovation and Infrastructure

Original Formulation: “Build resilient infrastructure, promote inclusive and sustainable industrialization, and foster innovation.”

- **Symmetry Integrity:** *Moderate.* Infrastructure and industrialization efforts often favor regions with existing capacity – typically urban centers or economically active zones – which can leave less-developed areas further behind. An adversary might funnel infrastructure investment to politically loyal regions or high-return projects, neglecting poorer or remote communities (thus widening regional disparities). For instance, building highways and tech hubs in the capital while villages remain without roads or internet fails the “inclusive” aspect. Similarly, innovation initiatives could concentrate on sectors that benefit an educated few, not broad society. If not carefully managed, Goal 9 could reinforce a familiar asymmetry: modern infrastructure for the “haves” (industrial zones, cities) vs. minimal improvements for the “have-nots” (rural poor). Additionally, global asymmetry can manifest: wealthy countries might dominate innovation and reap most benefits, while poorer ones struggle to industrialize under global competition, yet aggregate statistics could still show some global progress. That said, good infrastructure inherently has potential to reduce inequalities if distributed fairly – the key is whether an adversarial approach biases that distribution.
- **Incentive Integrity:** *Moderate.* This goal combines several elements, which could lead to **cherry-picking incentives**. For example, it's easier to build big infrastructure (roads, bridges) than to ensure those investments are climate-resilient or benefit all. A government may chase the number of projects completed or dollars invested in infrastructure as a sign of progress, regardless of actual utility or sustainability. There's incentive to prioritize **visible, grand projects (sometimes called “white elephants”)** that look impressive (for political gain or corruption opportunities) rather than smaller, local infrastructure that might yield more inclusive benefits. Industrialization is often measured by manufacturing value-added or share of GDP; a country could boost those by establishing special economic zones or attracting a few large factories, even if those don't create broad employment or are foreign-owned enclaves with limited linkage

to local economy. So one can game industrialization by quantity but not quality (e.g., polluting industries that rack up output but offer poor jobs). Innovation is hardest to measure (maybe via R&D spending or number of patents); there's incentive to create superficial innovation programs or count patent filings that may not translate to real progress. In essence, actors might do the straightforward parts (build infrastructure, raise industrial output) but ignore inclusivity and sustainability if those complicate the immediate metrics.

- **Measurement Hardness:** *Moderate-High (for some sub-areas).* Infrastructure can be quantified (e.g., km of roads, broadband penetration), and industrial output is captured in economic data. Those are “hard” numbers, albeit not telling the whole story. Innovation is softer; while one can count patents or R&D spending, quality and impact of innovation are tough to gauge. The composite nature of Goal 9 means **progress is measured through proxies**: we infer resilience perhaps by infrastructure investment in relation to GDP, or inclusivity by share of small-scale industries, etc., but these are indirect. As a result, a government can tout certain metrics (like “we doubled our road network”) while neglecting others (like “how many people gained access or how maintained it is”). Also, infrastructure quality and resilience can be hidden issues until tested (e.g., construction standards skimmed via corruption won't show until a disaster strikes). So an adversary might skimp on quality to hit quantitative targets. The sustainability of industrialization (emissions per unit of output) might not be fully captured either, allowing pollution to be an unseen side effect. In summary, plenty of room exists to present a rosy picture via selective indicators without delivering truly resilient, inclusive systems.
- **Capture Resistance:** *Low.* Infrastructure and industry are classic areas for **rent-seeking and capture by elites**. Big construction and industrial projects are often driven by powerful conglomerates or require government contracts – fertile ground for corruption or favoritism. An oligarch or political cronies might secure contracts for infrastructure (possibly inflating costs or building unnecessary projects for profit). The goal could be perverted into a crony enrichment scheme under the guise of development. Industrial policy can be captured by incumbent industries lobbying for protection or subsidies (for example, fossil fuel or heavy industries might lobby to be labeled “sustainable industrialization” recipients). Innovation efforts could be captured by well-connected firms seeking research grants or by a narrative that favors a certain technology path that benefits a few players. Internationally, infrastructure finance (like through development banks or China's Belt & Road) might align with geostrategic interests rather than purely local needs, effectively capturing Goal 9 to advance donor countries' agendas. Without strong anti-corruption and competitive procurement practices, this goal is a playground for graft. Also, choices of infrastructure have long-term lock-in; an adversarial actor could push infrastructure that locks dependence on them (e.g., proprietary tech or debt traps), thereby capturing future economic levers.
- **Trust Degradation Risk:** *Moderate.* People often judge development by visible infrastructure; if new roads, ports, and networks appear and create jobs, it can bolster trust in progress. However, **failed or misused infrastructure has the opposite effect**. Empty or collapsing projects (the notorious ghost infrastructure) become symbols of waste and broken promises. If industrialization efforts lead to environmental harm (polluted cities, unsafe factories) or displacement (land taken for projects without fair compensation), public trust erodes, especially among affected communities. Also, if the benefits of infrastructure/industry seem to accrue only to the rich or foreigners, citizens may become resentful (“They built a fancy highway but I still don't have transport to my village”). Innovation can also be polarizing – for example, automation or modern industries might leave older workers jobless, causing them to distrust the narrative of “innovation is good for all.” Should corruption scandals emerge (and infrastructure sector is notorious for this), trust in government and institutions declines sharply. In an adversarial spin,

opposition figures will highlight white elephant projects and graft to discredit the ruling establishment and by extension the multilateral goals it aligned with. Overall, Goal 9's impact on trust depends on execution: success stories build legitimacy, but poor implementation or clear inequities in who benefits can deeply undermine public confidence in development initiatives.

Survivability Score: 4/10. *While infrastructure and industry deliver tangible benefits, the sector's vulnerability to corruption and unequal distribution is high. An adversarial environment could see many projects that nominally fulfill Goal 9 but actually entrench elite interests or produce marginal real improvements. The goal's broadness (resilience, inclusion, innovation all in one) makes it easy to claim partial success and gloss over failures. Without accountability, the physical legacy could be a mix of useful assets and costly boondoggles – the latter fueling cynicism. Thus, the goal's survivability under hostile conditions is on the low side.*

Goal 10: Reduced Inequality

Original Formulation: “Reduce inequality within and among countries.”

- **Symmetry Integrity:** *High in intent, Low in practice.* This goal explicitly targets symmetry – both within societies (equality of outcomes or opportunities regardless of income group, gender, ethnicity, etc.) and globally (narrowing the gap between rich and poor nations). As such, its very *raison d'être* is to correct asymmetries. However, in practice, **it's one of the hardest goals and is highly susceptible to power dynamics.** Domestically, those with wealth and privilege often resist redistributive policies; an adversary in power may give lip service to equality but implement minimal changes or even policies that increase inequality while using clever framing (e.g., tax cuts labeled as pro-growth). Internationally, powerful countries might not actively cooperate to reduce global inequities (for instance, on trade regimes, debt relief, or migration). So while symmetry is the core, the adversarial forces around it are strong – symmetrical outcomes threaten entrenched interests. Without strong commitment, one can declare support for Goal 10 yet do nothing that truly shifts the balance, preserving asymmetry. Indeed, critics have pointed out that **the SDGs rely on voluntary national actions and do not strongly constrain wealthy actors**, thus possibly *exacerbating global inequality* because richer entities can continue on course ² .

- **Incentive Integrity:** *Low.* For those in power (national elites or wealthy states), there are **weak incentives and often strong disincentives** to meaningfully reduce inequality. Metrics for inequality in SDG 10 (such as growth of bottom 40% incomes vs national average, or Gini coefficient) might be improved through superficial means – e.g., a short-term cash transfer or raising minimum wage slightly – without altering structural inequities (like wealth concentration, monopoly power, land ownership, etc.). A government could focus on improving the lot of the “middle class” and claim inequality is addressed while the top 1% still take a huge share. Since the goal is broad (covering inequality due to income, opportunity, and other dimensions), an adversary can cherry-pick one aspect to show progress (for instance, increase education access for the poor, which is good but doesn't immediately change income distribution) and downplay stagnant wages or regressive taxation. Internationally, a rich country might highlight its aid increases or some debt forgiveness (small relative to its wealth) while maintaining larger systemic advantages. The goal has no binding mechanism to compel sharing of wealth/power; thus an adversary with means might publicly endorse fairness but privately prioritize maintaining their advantage. In summary, it's easy to pay lip service and make token efforts that slightly move an indicator, without truly threatening status quo inequality – which would be the adversary's preference.

- **Measurement Hardness:** *Moderate.* Income inequality is quantifiable (Gini, Palma ratio, etc.), and data exists, though with lags. However, inequality has many facets: wealth inequality (which is often more extreme) is less tracked; inequalities of opportunity (education, healthcare access) need proxies. Also, inequality *among* countries is usually measured by things like per capita GDP differences, which can shrink if large developing countries grow faster – but that doesn’t mean poorer nations’ people are catching up in absolute terms. There’s a technical nuance: global inequality might decline if a few populous countries (e.g., China, India) reduce poverty, even while within-country inequality surges – one could tout the former and ignore the latter. A savvy actor might use aggregate global stats to claim progress that doesn’t reflect local realities (e.g., “global inequality is down” thanks to some convergence, while their own country’s inequality worsened). Also, inequality data can be politically sensitive; governments may suppress or cast doubt on figures (e.g., not conduct household surveys that reveal extreme concentration at the top). Many countries lack real-time data on top incomes since it requires tax data which may not be public. So, a regime could under-report the share of wealth of elites, or simply not report on inequality indicators, dodging scrutiny. Without consistent, high-quality data, advancing or monitoring this goal is difficult, giving adversaries cover to declare commitments but claim “we lack data to assess yet.”
- **Capture Resistance:** *Very Low.* By definition, efforts to reduce inequality challenge those at the top, who have the means to **capture the political and economic system** to protect their interests. Domestic elites can influence policy (via lobbying, funding political campaigns, controlling media narratives) to dilute any redistributive reforms. For instance, proposals like higher taxes on the rich, land redistribution, stronger labor rights, or anti-monopoly laws may be stalled or watered down by vested interests. They might push more “acceptable” inequality fixes such as charity or minimal social programs that don’t truly threaten their wealth. On the international stage, wealthy nations and corporations often dominate trade negotiations and global finance rules – they can resist changes that would, say, give poorer nations more say in IMF/World Bank or restructure trade in favor of developing industries. There is also intellectual capture: economic paradigms that downplay inequality concerns can be promoted by elite-controlled institutions, framing the discourse such that inequality is seen as secondary or inevitable. Essentially, the **fox often guards the henhouse** with Goal 10 – those who would need to relinquish some advantage to meet it can exert power to ensure they don’t have to. This makes it one of the goals most susceptible to co-option or paralysis.
- **Trust Degradation Risk:** *High.* Inequality is palpable to people; when disparities grow or persist, it breeds frustration and erodes trust in governance. If SDG 10 is not met or is seen as a sham (lots of talk, little actual leveling), it can validate cynics who say global initiatives only protect the rich. Already, post-2015 years have seen rising discontent about inequality (fueling protests, populism). Should the public perceive that the UN goals allowed the rich to get richer (which in fact happened in recent years, exacerbated by events like the pandemic) ²⁰, confidence in the fairness of the international order diminishes. Moreover, focusing on inequality “among countries” could even be misused by nationalists: e.g., a powerful country might claim “we must grow faster than others to not fall behind” – twisting the goal to justify competitive or protectionist policies, which could sow distrust between nations instead of solidarity. Conversely, if international cooperation on inequality (like tax haven crackdowns or equitable vaccine distribution) fails, poorer nations will distrust promises of a just global partnership (a cross with Goal 17). In short, failure on Goal 10 – which many consider a litmus test for whether the SDGs truly benefit everyone or just the elites – would sharply reduce trust in the Agenda 2030 project. It could lead to public apathy (“nothing changes for the common man”) or anger (“system is rigged”), both corrosive to social cohesion and global cooperation.

Survivability Score: 2/10. *Goal 10 is arguably the most politically challenging and thus least likely to survive adversarial pressures intact. The incentives of powerful actors run counter to it, data can be obfuscated, and genuine progress requires tackling deep-rooted power imbalances – something adversaries will actively resist. Without enforceable mechanisms or strong social movements forcing change, this goal is in peril of being the least fulfilled. Its low score reflects that it's the easiest to pay lip service to while undermining in substance, and such betrayal would not go unnoticed, striking at the heart of SDG credibility.*

Goal 11: Sustainable Cities and Communities

Original Formulation: “Make cities and human settlements inclusive, safe, resilient, and sustainable.”

- **Symmetry Integrity:** *Moderate.* Goal 11 addresses urban and community development, with an emphasis on inclusivity – e.g., upgrading slums, accessible transport, green public spaces for all. In a fair scenario, improvements benefit rich and poor neighborhoods alike. However, urban development often produces **spatial inequalities**. An adversarial actor might focus on high-profile city center projects while neglecting slums or peri-urban areas where the poor live, thereby *worsening* spatial asymmetry. Gentrification is another issue: “urban renewal” might actually push low-income residents out (making a city more “sustainable” or modern-looking, but less inclusive). Thus a city can become greener and safer for the affluent while the poor are relocated to peripheries or informal settlements remain squalid. Globally, there's also imbalance between well-resourced cities in developed countries and rapidly growing cities in developing countries; the latter may struggle without significant support. If global efforts don't address these disparities (e.g., via knowledge or funding transfer), the gap in urban liveability could widen. So while symmetry is a key intention (inclusive for all residents), without explicit guardrails an adversary might prioritize cosmetic sustainability (like a new park downtown) at the cost of equity (ignoring that many lack basic services).
- **Incentive Integrity:** *Moderate-Low.* Urban metrics include things like proportion of urban population living in slums, access to public transport, air quality, disaster plans, etc. These can be gamed. A city government might choose to redefine boundaries or what counts as a slum to report better numbers (for instance, reclassifying an area so it's not officially a “slum” even if conditions haven't improved). Or there may be an incentive to do quick fixes that improve an indicator but not lasting solutions – e.g., relocating some slum dwellers temporarily to bring down the percentage in informal housing, without ensuring they have permanent affordable homes. In transport, it's easier to build a few big transit lines than to ensure **affordable last-mile connectivity**; a government could boast a shiny metro system (helps the indicator of share of population with convenient transit access) yet many poor still can't afford fares or reach stations. For safety and resilience, an administration might create plans on paper (disaster risk reduction plans, etc.) to tick the box, but not actually invest in implementation. There's also a tendency to favor projects that are visible (smart city tech, big real-estate developments) rather than less glamorous work like improving drainage in informal settlements – an adversarial city leader might chase the former to attract investment or prestige, claiming it as “innovation for sustainability,” while leaving fundamental inclusivity issues unaddressed.
- **Measurement Hardness:** *Moderate.* Some aspects (housing quality, transport coverage) are measurable but data can be inconsistent. For example, what qualifies as “adequate housing” or a slum upgrade is somewhat subjective. Numbers on access to services in cities might not capture disparities within a city. Air quality is measurable continuously, but an adversary could cite city-wide average improvements which may come from moving polluting industry out (perhaps to another area), not overall reduction – effectively exporting the problem. Indices of sustainability for cities exist but often aggregate many factors into a composite score that can be manipulated

by focusing on easy gains. Also, cities are complex systems with multiple jurisdictions (municipal, metropolitan, etc.), so reporting can be fragmented or cherry-picked. If one district is doing well, a city might highlight that and omit worse areas in reports. The word “communities” implies rural settlements too, but many countries treat Goal 11 as urban-only, potentially ignoring rural needs (assuming other goals cover those). Thus measurement might skip smaller communities entirely. Finally, resilience (to disasters, climate) is only proven when a shock hits; until then, one might assume resilience on the basis of having a plan or infrastructure that hasn’t been tested – giving a false sense of security if those plans are never implemented beyond paper.

- **Capture Resistance:** *Low-Moderate.* Urban development is often swayed by real estate developers, construction firms, and political interests. City planning can be captured by those who stand to profit from land development – for instance, developers pushing luxury projects that consume public space or displace low-income housing, all while being marketed as “modernizing the city.” Municipal corruption or collusion is a known issue; zoning laws can be bent to favor certain projects. The sustainability theme could be co-opted by companies selling “smart city” solutions or high-tech infrastructure, which may not address basic community needs but fit a narrative of innovation. Also, local governments might prioritize projects that gain them political favor (like visible beautification before elections) over long-term inclusive planning. At a higher scale, international investors might influence city agendas (e.g., through infrastructure funds or banks financing toll highways vs. public transit). Unless governance is transparent and participatory, capturing Goal 11’s implementation to benefit specific economic actors is quite feasible. The ones left out – often the urban poor – have less voice, so their interests can be ignored despite the “inclusive” mantra.

- **Trust Degradation Risk:** *Moderate.* City residents experience daily whether things are improving: housing, traffic, pollution, safety. If SDG-related initiatives improve these in an equitable way, it’s very tangible and builds trust in local authorities and by extension the idea of global goals. Conversely, if policies are seen as catering to the wealthy (e.g., evicting street vendors to make a city “cleaner,” or building green spaces that only gentrify neighborhoods), marginalized groups will feel betrayed and angry. Urban protests often ignite around such issues (evictions, transit fare hikes, pollution crises). A failure to involve communities in planning can lead to backlash, undermining trust in both local governance and any external partners like development agencies. Also, a disaster like a flood or building collapse, if attributed to ignored warnings or poor infrastructure, can rapidly erode public confidence – people might question all the talk of “resilience” if actual preparedness was lacking. On a global narrative level, if cities remain extremely unequal and unlivable for many by 2030, it calls into question the success of the SDGs where most of humanity now lives (cities). That said, trust impacts might remain localized – people blame their mayor or government more than the UN if their city fails, but it still feeds into overall skepticism about grand promises.

Survivability Score: 5/10. *Goal 11 is a mixed bag. Many cities are taking positive steps and sharing best practices, giving hope that the goal can weather some adversarial challenges. However, urban development’s political economy – high land values, vested interests – means the goal can be undermined by those pursuing profit over inclusivity. Our moderate score reflects that outcomes will likely be very uneven: some cities will achieve great strides (increasing resilience and inclusion), boosting the narrative, while others will showcase inequality or botched projects, harming credibility. On average, the goal’s implementation is only somewhat robust to adversarial influences, requiring constant civic vigilance to truly be “for all.”*

Goal 12: Responsible Consumption and Production

Original Formulation: “Ensure sustainable consumption and production patterns.”

- **Symmetry Integrity:** *Moderate-Low.* This goal implicitly challenges how resources and benefits are distributed and used across the world. There is a global symmetry issue: high-consuming countries vs. low-consuming ones. A fair interpretation would mean the rich world moderates its consumption footprint and helps poorer regions develop sustainably. However, adversarially, wealthy countries might externalize the problem – e.g. continue high consumption while outsourcing dirty production to developing countries, then claim they reduced domestic emissions (looking “responsible” at home but shifting impacts abroad). That leaves global asymmetry intact or worse: the developing regions bear the pollution and resource depletion to supply products for the developed world. Within countries, if measures are taken, they might affect groups differently – e.g., environmental regulations could raise prices that burden the poor more, or campaigns focus on individual behavior of consumers while letting big industries off the hook (an asymmetry in accountability). True symmetric responsibility would scale by capacity and impact, but adversaries can skew it so the less powerful end up constrained while powerful actors maintain lifestyles. So, symmetry can be subverted by making the “weak” do more (or suffer more consequences) and letting the “strong” off lightly.
- **Incentive Integrity:** *Low.* Achieving sustainable consumption/production often means altering business models and consumer habits in ways that incumbents resist. There are **strong incentives to greenwash rather than change fundamentally**. Companies might adopt voluntary sustainability pledges or produce “eco-friendly” product lines that barely scratch the surface, all to claim alignment with Goal 12 while continuing core unsustainable practices. For example, a plastic producer could support recycling programs (small investment) instead of reducing plastic output (which hits profit). Governments have incentive to show they are doing something via recycling rates or banning some single-use plastics, which are visible actions, rather than tackling deeper issues like overconsumption, wasteful supply chains, or fossil fuel subsidies. Consumption and production patterns are hard to regulate without affecting economic growth and powerful lobbies; thus the path of least resistance is to promote consumer awareness campaigns (shifting onus to individuals) and voluntary corporate responsibility, which often yield modest results. An adversary might also game metrics: e.g., count the number of companies reporting on sustainability (an output) rather than actual reduction in environmental footprint. There’s also a loophole in international accounting: a country can say its domestic material consumption or carbon footprint improved, but that might be by importing goods (so the impacts occur elsewhere). Such incentives mean each actor tries to look good individually without addressing systemic global throughput – effectively a tragedy-of-commons scenario where everyone hopes others will consume less while they continue as usual.
- **Measurement Hardness:** *High.* This goal is extremely broad – it covers use of natural resources, waste generation, recycling, corporate sustainability practices, public procurement, etc. Indicators range from material footprint per capita to food waste per capita to national sustainable development policies. Many of these are difficult to measure accurately or compare. For instance, **material footprint** tries to allocate global resource extraction to final consumers, which requires complex modeling and is not regularly tracked in detail by all countries. Food waste data relies on estimates with high uncertainty. Corporate sustainability reports (another aspect) are self-reported and not standardized, easily allowing selective disclosure. With such soft metrics, an adversary can cherry-pick improvements (e.g., “we reduced domestic material use by X%” ignoring that imports rose). Waste and recycling figures can be manipulated (sending waste offshore might remove it from your stats, classifying waste in certain ways can change

numbers). There's also often a time lag and lack of transparency in supply chains, so consumers and regulators may not know the true footprint of products. In essence, measuring "sustainability" of consumption/production is so complex that it's easy to obscure lack of progress or even regress behind data gaps and accounting tricks ¹² ³. This opaqueness benefits adversarial actors who might loudly report narrow successes (like "50% of our paper is recycled now!") while ignoring the bigger picture (overall deforestation still rising, etc.).

- **Capture Resistance:** *Low.* Corporations and affluent consumer markets clearly have a huge stake in how this goal is shaped. **Capture by corporate interests** is a prime threat: industries may shape "sustainable production" standards that are weak (to avoid real change) but give the appearance of responsibility (e.g., voluntary certification schemes with lax criteria). The goal's implementation often relies on multi-stakeholder partnerships – these can be captured by industry to avoid hard regulation. For instance, the fashion or tech industry might promote recycling initiatives which they manage, preempting stricter laws on waste or sourcing. Advertising and media (influenced by corporate sponsors) might dilute messaging, turning sustainability into a marketing buzzword rather than a call for reduced consumption. Rich countries could capture the narrative by pushing solutions that benefit their firms (like selling high-tech "green" products to solve sustainability, rather than addressing overconsumption levels). The fossil fuel industry might capture discussions on sustainable energy use by diverting focus to consumer behavior (e.g., "turn off your lights" campaigns) away from systemic changes (like ending oil extraction). Essentially, this goal threatens established business models if taken seriously, so capture efforts are to be expected – steering it towards incremental, non-threatening actions (making it about recycling and efficiency rather than consuming less or redistributing resources). Without strong independent voices (scientists, NGOs) holding the line, the vested interests can water down the ambition significantly.
- **Trust Degradation Risk:** *Moderate.* People are increasingly aware of environmental limits and corporate greenwashing, so if Goal 12 results in a lot of rhetoric but continued environmental degradation, it will foster cynicism. Already, decades of talk about sustainable development have seen consumption and emissions rise, which leads many (especially youth) to doubt the sincerity of governments and businesses. Should 2030 arrive with little change in lifestyles of the rich or in global resource depletion, trust in global frameworks to solve these collective action problems will be undermined. On the other hand, if radical measures were taken without public buy-in (like sudden consumption taxes or removing popular products), that could also spark backlash and distrust ("the UN agenda is forcing us to change our way of life!" – a narrative some conspiracy theorists already push). The challenge is balancing earnest progress with keeping the public onboard; if adversarial forces tip that balance (either by doing nothing or by imposing clumsy measures), trust suffers. Moreover, failure on this goal has cross-cutting implications: it connects to climate, biodiversity, etc. A perceived failure here might be seen as failure of the whole sustainability project. If people observe that corporations still pollute freely and consumers in wealthy areas haven't adjusted their habits at all, they may conclude the SDGs were toothless – just feel-good talk while the planet's problems intensify ²¹. That erosion of trust could then hamper future collective efforts (like public support for climate policies or belief in scientific warnings).

Survivability Score: 3/10. *Goal 12 is highly vulnerable because it asks for deep changes in entrenched patterns with relatively weak enforcement. Adversarially, it's easy to subvert via token actions and narrative control. Many advanced economies are off-track on this goal (resource use and waste still rising), and powerful interests prefer it that way. Unless strong accountability is introduced, this goal will likely see minimal genuine progress, making it one of the least robust SDGs against adversarial forces. The low score*

reflects that – under status quo incentives – “responsible consumption/production” might largely remain an aspirational slogan rather than a realized shift by 2030.

Goal 13: Climate Action

Original Formulation: “Take urgent action to combat climate change and its impacts.”

- **Symmetry Integrity:** *Low-Moderate.* Climate change is a quintessential symmetry problem: those who have historically emitted the most are not the ones who suffer first or most; impacts hit poorer countries disproportionately, even though their emissions are small. The UN climate framework acknowledges common but differentiated responsibilities, but in practice, **power asymmetries skew the response.** Wealthy countries might set inadequate targets and still not meet them, while poorer countries face pressure to curb future emissions or are left to deal with adaptation largely on their own. An adversarial stance from high emitters (past or present) might be to insist others (developing economies) also take equal action despite unequal capacity, or to use climate aid as leverage rather than genuine support. Within countries, there's asymmetry too: vulnerable communities (coastal, indigenous, low-income) bear the brunt of climate impacts and sometimes of climate policies (e.g., carbon pricing can hit the poor harder if not designed carefully). If adversaries shape policies, they may protect certain industries or regions at the expense of others – e.g., keep coal jobs (local benefit) while global climate suffers, or shift to renewables in one area but dump fossil projects on marginalized areas. True symmetry (both in effort and protection) is hard to achieve and easy to dodge; we often see a pattern where those with more resources insulate themselves or delay action, leaving others more exposed, contrary to the fairness principle.
- **Incentive Integrity:** *Low.* The incentives to cheat or free-ride on climate action are notorious. Reducing emissions often has short-term economic costs but provides global long-term benefits, so every actor has an incentive to let others do more. Despite growing urgency, **national interests often override:** e.g., resource-rich countries face incentives to continue exporting fossil fuels; politicians have incentives to favor immediate economic growth over long-term climate goals (especially given election cycles). Even within the SDG context, Goal 13's phrasing is vague (“take action”) and mostly refers to aligning with the Paris Agreement, which relies on nationally determined contributions (NDCs) – essentially voluntary pledges. An adversarial country could set a weak NDC, or one with loopholes (like relying heavily on offsets or future technologies), appearing compliant but not actually decarbonizing much. There are also creative accounting tactics: using favorable baselines, excluding certain sectors, or counting reduction in deforestation as climate action in one place while ignoring new drilling elsewhere. Corporations might lobby for flexibility, such as being able to use carbon credits of dubious quality to claim “net-zero” while continuing to emit ¹³. These behaviors reflect incentives to do the minimum while claiming the maximum credit. Without an enforcement mechanism, it's rational (though unethical) for actors to promise distant “net-zero by 2050” and do little now. Climate finance promises (to support developing countries) similarly suffer from incentives to over-report and under-deliver, since donor governments get praise for pledges but face little consequence for not disbursing funds fully or counting existing aid as “climate finance.”
- **Measurement Hardness:** *Moderate.* Emissions data is actually one of the more quantifiable aspects – CO₂ can be measured fairly well at national levels (though there are uncertainties and reporting delays). However, the *quality* of action is harder to measure. One can tally policies adopted or climate plans made, but assessing their adequacy or enforcement is complex. A country could have a climate adaptation plan (an indicator for Goal 13) on paper but not implement it; measuring actual resilience is only evident when a disaster hits (lagging indicator).

Similarly, mitigation efforts are measured by emissions trends, but short-term variability (like an economic downturn reducing emissions) can mislead. Countries may claim credit for reductions that were due to unrelated factors (e.g., outsourcing heavy industry). And some aspects, like education/awareness on climate (one of the targets), are qualitative. At the global aggregate, we can measure atmospheric GHG concentrations; adversarially, even if that keeps rising (indicating failure), individual actors might point to their own reported efforts to deflect blame. Because climate change is global, measurement of one actor's progress needs context of global outcome – which currently is not good (we're off-track for the Paris goals). But each can say "not my fault." Also, climate impact data (extreme events, losses) is measurable yet attributing cause and response is tricky; an opponent might downplay climate links or highlight adaptation successes to mask overall risk growth.

- **Capture Resistance: Low.** The climate policy arena is heavily contested by powerful interests, particularly the fossil fuel industry and high-emitting sectors (transport, heavy manufacturing, industrial agriculture). These actors have a long history of lobbying, funding misinformation, and influencing policy to delay or dilute climate action. Goal 13 implementation can be captured by those interests ensuring that solutions favor them – for example, promoting carbon capture and storage (thus extending oil and gas use) or pushing natural gas as a "bridge fuel" as official climate strategy. Some countries' delegations to climate negotiations are deeply tied to their fossil industries (either state-owned or private), leading to captured national positions. Even adaptation funding can be captured – e.g., contractors and companies positioning to get major infrastructure projects (sea walls, etc.) rather than community-based solutions. There's also risk of capture by geopolitical interests: countries might trade climate commitments for concessions elsewhere, treat it as bargaining chip rather than an urgent goal. On the information side, narrative capture is rampant – framing climate action as too costly or infringing on sovereignty, which slows mobilization. While the global scientific consensus is strong (less susceptible to capture), translating it to policy has been stymied by these influences. The presence of fossil fuel lobbyists at climate summits and certain countries watering down language in UN texts are clear examples of capture attempts.
- **Trust Degradation Risk: High.** Climate change is the defining issue for many, especially the younger generation. Repeated failure to act decisively (as we've seen since the 1990s) has already eroded trust in political leaders and international processes. If Goal 13 – with its call for "urgent action" – is not met, or if warming continues unabated, public trust in institutions can plummet. It might manifest as climate anxiety turning into anger, protests, or disillusionment ("They talked a lot but sacrificed our future"). In fact, progress is so off-track that this scenario is very plausible without drastic change. Adversaries (like populists) might exploit either side: if strong climate policies are implemented, they may rally those inconvenienced (e.g., "the elites care more about carbon than your jobs" narrative), undermining trust in climate science and institutions. Or if climate disasters strike and governments are unprepared, trust collapses due to perceived negligence. Already, segments of society believe climate goals are part of some sinister agenda (conspiracy theories) or conversely that governments are simply bought by oil companies – both reflect low trust. The UN and governments stake credibility on addressing existential threats; failure to do so by 2030 will be a glaring indictment. Conversely, if by some coalition effort climate action ramps up in time, it could restore faith. But as things stand, this goal's success is very much in question and thus a huge reputational risk for the whole SDG enterprise.

Survivability Score: 4/10. *Climate Action is a paradox: scientifically, we know what to do, and there's global attention, but politically it's fiercely contested. The goal's vague wording, reliance on voluntary pledges, and high stakes make it susceptible to delay and dilution. It scores somewhat low because adversarial forces*

(economic interests, nationalist politics) have significantly impeded progress, though not without pushback (civil society, some proactive coalitions). We give a slightly below-mid score acknowledging that while some advancement occurs (renewables growth, etc.), it's nowhere near the transformative scale needed. In an adversarial world, climate action often loses to short-term interests, threatening the goal's viability.

Goal 14: Life Below Water

Original Formulation: “Conserve and sustainably use the oceans, seas, and marine resources for sustainable development.”

- **Symmetry Integrity:** *Low-Moderate.* The oceans are a global commons, but not all benefit or suffer equally. Countries with large coastlines or advanced fishing fleets exploit marine resources heavily, while small island and coastal communities often bear the brunt of overfishing, pollution, and climate-related changes (like sea level rise, coral bleaching) despite contributing little to the cause. An adversarial dynamic is seen in international fisheries: powerful nations or companies fish in distant waters, often at the expense of local small-scale fishers. If Goal 14 is pursued weakly, asymmetry persists – wealthy nations might enforce sustainability in their own waters but continue unsustainable practices elsewhere (through fleets or imports). Pollution, like plastic and toxic waste, often comes disproportionately from some regions/industries but washes up as a problem for everyone, including those not at fault. Symmetry would require fair access and burden-sharing (e.g., limiting rich countries’ take, supporting poorer ones in conservation, sharing benefits like technology or marine genetic resources). Without that, an adversary can take a beggar-thy-neighbor approach: exploit what they can before others do, or free-ride on others’ conservation. Internally, the fishing industry dynamics often favor large trawlers over local fishers, again an asymmetry of power within countries that can persist if not addressed.
- **Incentive Integrity:** *Low.* The ocean suffers from **tragedy-of-commons incentives**: each actor (be it nation or company) gains by exploiting as much as possible and loses if they restrain themselves while others don’t. Illegal, unreported, and unregulated (IUU) fishing is lucrative and hard to police, so there’s incentive to overfish beyond quotas. Even regulated quotas often exceed sustainable levels due to industry pressure – a regulator might set a higher catch limit to appease fishers, assuming short-term gain and hoping stocks recover later (or that it becomes someone else’s problem). Pollution-wise, dumping waste into rivers/oceans is the cheapest disposal method for many industries (textiles, plastics, shipping); enforcement at sea is weak, so incentive to externalize costs there. Many marine conservation measures, like creating marine protected areas (MPAs), face incentives problems – a country might declare MPAs (easy announcement for SDG indicator) but not enforce them (hard, costly) or place them in areas that don’t inconvenience its industry (so minimal ecological benefit). The goal’s language “sustainably use” can be interpreted loosely; adversaries might lobby to define “sustainable” in a way that just repackages status quo fishing with minor tweaks. Climate change mitigation at sea (like reducing shipping emissions) is also slower because of cost incentives to continue cheap bunker fuel use. In short, without strong collective agreements and compliance, the rational (if unethical) move for many is to keep exploiting the ocean while it lasts, undermining Goal 14 severely.
- **Measurement Hardness:** *Moderate-High.* We can measure some things like fish stock status (percent overfished vs. sustainable), area of protected ocean, or acidification levels. But vast areas of ocean are data-poor. For example, fish stock assessments are expensive and cover only some commercial species; many fisheries in the developing world lack regular stock surveys, giving room to overfish unnoticed. Plastic pollution is measured via samples and modeling, but

those can be gamed by cleanup efforts in certain areas while production of plastic continues rising. Countries self-report their protected areas or policies; an adversary might inflate these by counting “paper parks” (MPAs that exist on maps but allow continued extractive activities). Also, the ocean is interconnected – one country’s actions affect another’s waters – yet the SDG monitoring is country-based, potentially missing that interplay. Tracking IUU fishing or deep-sea mining impacts is difficult, which adversaries exploit by operating out of sight (e.g., using flags of convenience or avoiding monitoring systems). With climate impacts like ocean temperature and pH, we rely on global data (which shows worrying trends), but attributing failures to specific actors is complex, letting everyone point fingers. Thus, measuring outcomes is possible and indicates problems (overfishing, reef loss), but measuring each actor’s performance and compliance is much tougher, something adversaries can hide behind.

- **Capture Resistance:** *Low.* Industries like commercial fishing (especially large fleets), deep-sea mining, shipping, and coastal developers have significant sway over marine policies. They often capture regulatory bodies or international negotiations to avoid strict limits on their activities. For instance, regional fisheries management organizations (RFMOs) sometimes get dominated by the interests of big fishing nations, making quotas too high or enforcement lax. In domestic contexts, politicians might side with local fishing lobbies or companies because of jobs and profits, even if scientists advise cuts. New industries such as seabed mining are pushing to set rules favorable to their future operations before strong environmental protections can be put in place. At negotiations like for a high seas treaty or plastic waste agreements, corporate lobbies (e.g., petrochemical for plastics, shipping industry) attempt to water down commitments. Essentially, those extracting wealth from the ocean will try to capture Goal 14’s implementation to ensure any “sustainability” measures don’t severely curtail their business. There’s also geopolitical capture: navies and nations see strategic interests in the seas (territorial claims, etc.), which can overshadow conservation concerns. For example, marine protected areas might be resisted if they conflict with potential oil drilling zones or naval routes. Without transparent, science-based, and inclusive governance, these forces can largely dictate outcomes.
- **Trust Degradation Risk:** *Moderate.* Marine issues resonate with the public when crisis points occur – e.g., images of plastic-choked whales or a collapsed fishery impacting local communities. If people see continued destruction of marine life despite lofty global goals, it feeds the narrative that governments and corporations can’t be trusted with stewardship of global commons. Coastal communities, in particular, might lose trust in national/international bodies if their livelihoods decline (fish catches dropping, coral reefs dying) while hearing promises of “sustainability.” Youth and environmental groups pay attention to things like ocean plastic; failure to significantly reduce it by 2030 would appear as a major broken promise (since it’s a very visible, tangible problem with much publicized attention). On the international stage, small island developing states (SIDS) and others dependent on the ocean could lose faith in multilateral processes if, say, rich nations block or slow a strong ocean treaty or fail to provide funding for marine conservation and climate adaptation that they’ve been calling for. However, compared to climate, ocean issues are a bit more out-of-sight for the global majority, so trust erosion might be more localized or sector-specific (e.g., fishers vs government) unless a dramatic event (like a fisheries collapse leading to global food price spikes or a charismatic species extinction) catches global attention. Still, as part of the environmental agenda, failing oceans contribute to the general cynicism about humanity’s ability to manage shared resources responsibly.

Survivability Score: 4/10. *Goal 14 is undermined by strong incentives to overuse a common resource and by weak enforcement mechanisms. While there have been some wins (like progress toward a high seas treaty, more MPAs, awareness of plastic pollution), these are often outpaced by ongoing degradation. The adversarial*

analysis suggests that unless global cooperation markedly improves, ocean sustainability efforts will lag. We score it lowish because current trajectories for fisheries and marine health are worrying, indicating the goal in its current form is not robustly surviving self-interest-driven assaults. It is marginally better than the inequality goal perhaps, because there's less ideological opposition to ocean conservation – it's more about neglect and exploitation – but still in a fragile state.

Goal 15: Life on Land

Original Formulation: “Protect, restore and promote sustainable use of terrestrial ecosystems, sustainably manage forests, combat desertification, and halt and reverse land degradation and halt biodiversity loss.”

- **Symmetry Integrity:** *Moderate.* Terrestrial ecosystem protection often pits local/resource-dependent communities against global/common interests. Ideally, benefits of conservation (ecological services, climate regulation) are shared widely, while costs (restrictions on land use, opportunity cost of not exploiting resources) are borne by some, usually rural or indigenous communities and developing nations with rich biodiversity. Symmetry would require those communities/nations be compensated or supported (e.g., via payments for ecosystem services, or benefit-sharing from genetic resources). In an adversarial scenario, powerful actors might impose conservation in ways that **burden local people unfairly** (like creating protected areas without local consent or adequate alternatives, effectively an asymmetry of who pays for sustainability). Alternatively, powerful economic interests might exploit ecosystems (logging, mining) disregarding impacts on marginalized groups who rely on the land – another asymmetry of power. Globally, there's asymmetry where richer countries have largely deforested historically but ask poorer ones to preserve forests now (justified for climate/biodiversity), which can breed resentment if not coupled with support – adversaries can play up this “you industrialized, now you want to stop us” narrative. Also, biodiversity benefits (like new medicines) often accrue to companies in developed countries, whereas biodiversity-rich nations or indigenous communities might not see a fair share – a symmetry issue that has led to calls for access-and-benefit-sharing. Without mechanisms to address these, trust and participation in Goal 15 can be uneven.
- **Incentive Integrity:** *Low.* Economic incentives strongly favor land use that often conflicts with conservation: agriculture, logging, mining, infrastructure. **Short-term profits and development goals drive deforestation and land conversion.** Politicians have incentives to allow such activities for GDP growth, jobs, or personal gain (corruption), while conservation yields diffuse benefits and fewer immediate gains. Even well-intentioned governments may face pressure from constituencies (farmers wanting more land, industries wanting permits). The global market incentivizes producing commodities (cattle, soy, palm oil, timber, etc.) cheaply – often at the environment's expense – and consumer countries' demand fuels that. While there are pledges (like zero-deforestation supply chains), adversarial actors can bypass them via complex supply chains or simply move to markets with less scrutiny. Poaching and illegal wildlife trade continue because the potential profits (for rare animal parts, timber) are high and enforcement is weak – poachers and traffickers thus have incentive to risk it. Protected areas on paper might be understaffed or encroached upon because local officials see more benefit in allowing farming or logging. The goal tries to instill sustainable practices, but unless those are also profitable or heavily enforced, actors have incentive to do the unsustainable but lucrative thing. In sum, unless conservation is incentivized through funding or sustainable alternatives, adversaries (and even average actors) will lean toward land degradation for gain.
- **Measurement Hardness:** *Moderate.* Some things are measurable (forest cover change via satellites, number of species threatened via assessments, extent of protected areas). However,

biodiversity is complex and not fully captured by any single metric. A country might maintain forest cover but the quality changes (primary forest replaced by monoculture tree plantations – technically “forest” but ecologically not equivalent). If an indicator only counts area, this nuance is lost and can be gamed. Countries report on things like progress in sustainable forest management via composite indices, which involve self-assessment. That allows optimistic reporting. Land degradation (desertification) is sometimes assessed via proxies like vegetation index or soil erosion estimates, with uncertainties that could be swayed by natural variability (e.g., a good rainy year might temporarily “green” degraded lands). Also, data often lags; by the time deforestation is reported in official stats, a lot might have happened (though this is improving with near-real-time satellite monitoring initiatives). Another potential gap: focusing on protected area coverage (easy to measure % of land protected) could hide whether those areas are effectively managed or just “paper parks.” Similarly, species extinction or recovery is slow to detect; a nation might claim no extinctions on its watch, but species could be functionally extinct (numbers so low they can’t recover) yet not declared officially. Thus, adversaries can tout certain numbers (like “we have 25% of land under protection” or “we planted X million trees”) which sound good but might not equate to real biodiversity health if those trees are plantations or protection is nominal.

- **Capture Resistance:** *Moderate-Low.* Environmental policy can be captured by agribusiness, logging companies, mining interests, and corrupt officials. These groups often lobby to weaken protected areas (redraw boundaries, allow exemptions) or environmental regulations. We’ve seen cases of environmental agencies being headed by people with industry ties who then facilitate permits or reduce enforcement (“fox guarding the henhouse”). In some regions, powerful landowners resist land reform or indigenous land rights that could aid conservation. Internationally, negotiations on biodiversity funding or rules can be influenced by countries with interests in genetic resources or biotechnology (like debates on digital sequence information benefit-sharing, etc., where capture by commercial interests can shape outcomes). However, conservation also has strong NGOs and public support in many places, which can counter blatant capture to some degree. But in adversarial contexts, those voices might be marginalized or even persecuted (sadly, environmental defenders are often threatened or harmed). Another capture aspect: climate initiatives like carbon offsets could “capture” land use agendas – e.g., big companies buying land to plant monoculture forests for carbon credits, which might tick climate metrics but do little for biodiversity or even displace local communities. That’s a captured form of Goal 15 where corporate climate optics override holistic ecosystem needs. Without inclusive and transparent governance, these trends can compromise the goal’s integrity.
- **Trust Degradation Risk:** *Moderate.* People’s trust can wane if they see continuous nature destruction despite grand commitments. Locally, if communities are promised sustainable development but instead see their forests cut or land degrade, they lose trust in authorities and possibly turn to unsanctioned coping methods (like illegal logging to survive, since the system didn’t protect their interests). Internationally, biodiversity loss – like iconic species declines (elephants, tigers, etc.) or burning rainforests – garner wide public attention, and failure to address these by 2030 would make many question the efficacy of international agreements. There’s already a sense that biodiversity has been the neglected sibling of climate in terms of action; if that continues, faith in processes like the Convention on Biological Diversity might erode. Conversely, heavy-handed conservation (like fortress conservation that evicts people) can create distrust and backlash among local populations toward conservation authorities and goals. Indigenous and local communities may become cynical about global promises if their rights and knowledge are not respected in implementation of SDG 15. On the flip side, visible successes (like a forest cover increase or species recovery) can build trust – so the risk is more in unfulfilled promises or negative side-effects. Also, an undercurrent: as environmental crises intensify (e.g.,

dust storms from desertification, zoonotic disease outbreaks from ecosystem disruption), public trust in whether leaders are doing enough to safeguard the planet for future generations is at stake. If Goal 15 falls short, it contributes to a broader narrative of failed stewardship.

Survivability Score: 4/10. *Life on land is under heavy pressure from economic forces, and protective efforts, though growing, are often outpaced. This goal has many moving parts (forests, deserts, biodiversity), each with its own battles. Adversaries – from illegal loggers to profit-driven corporations – often have the upper hand unless strong governance and incentives align for conservation. The score reflects that currently deforestation and species loss continue at alarming rates, suggesting the goal is not on track and is vulnerable to being undermined. Without a paradigm shift (valuing natural capital properly, empowering local stewards, penalizing polluters), Goal 15 remains one of the more fragile objectives in the face of adversarial exploitation.*

Goal 16: Peace, Justice and Strong Institutions

Original Formulation: “Promote peaceful and inclusive societies for sustainable development, provide access to justice for all and build effective, accountable and inclusive institutions at all levels.”

- **Symmetry Integrity:** *Moderate.* Goal 16 covers a broad governance agenda, implicitly about leveling the playing field in society through rule of law and inclusivity. Symmetry here means equal treatment under the law, equal access to justice, and equal voice in institutions. Adversaries (especially authoritarian regimes or corrupt elites) often maintain asymmetrical systems – they might ensure peace for the privileged while using violence or coercion against marginalized groups, or justice that serves some (impunity for elites, harsh for the poor). If such an actor co-opts Goal 16, they might tout national “stability” (no open conflict) while ignoring state repression or exclusion of opposition and minorities. This goal can thus be subverted into a narrative of “law and order” rather than true inclusivity. Internationally, symmetry would entail all nations abiding by international law, and global governance giving equal weight to each voice, but powerful countries often bend rules (e.g., selective adherence to international courts or UN decisions) undermining the idea of equal justice among nations. Also, conflict and fragility often stem from historical asymmetries (inequalities, colonial legacies); addressing those is tough. If Goal 16 efforts don’t tackle root inequities (like by just propping up existing institutions that favor a group), superficial peace may hold but deeper justice isn’t served – an asymmetric calm.

- **Incentive Integrity:** *Low.* Governments or leaders who benefit from weak institutions, corruption, or oppression have clear disincentives to genuinely pursue this goal. **A repressive regime might want the veneer of SDG compliance** without loosening its grip – e.g., instituting a façade of anti-corruption drives that target opponents only, or creating “access to justice” through controlled courts that legitimize the regime rather than challenge it. There’s an incentive for such regimes to interpret Goal 16 narrowly: focus on certain crimes (like reducing petty crime for citizens’ sense of security) but not on, say, political freedoms or reducing torture. They might also manipulate crime or violence statistics to appear more peaceful (under-reporting conflict deaths, or classifying civil unrest as something else). For more democratic governments, the incentive may be to do the low-hanging fruit (e.g., e-government systems for some services to claim accountability wins) but avoid deeper reforms that threaten entrenched interests (like judicial independence or elite accountability). Internationally, powerful states have little incentive to reform global institutions to be more equitable (as that could reduce their influence). Aid for governance often comes with political strings, which can distort local incentives (e.g., passing laws to please donors, not necessarily to empower citizens). Also, because Goal 16 touches sensitive sovereignty issues, many countries might prefer soft commitments (like join a

partnership on transparency) but not hard ones (like allow external election observers or sign up to international justice mechanisms that could prosecute their officials). The presence of conflict can allow securitization to trump inclusion (a government could justify crackdowns in the name of peace). Overall, the incentives to partially implement are there (because peace and effective institutions are good in rhetoric), but full genuine implementation often threatens those in power – a strong disincentive.

- **Measurement Hardness:** *High.* Measuring peace, justice, and institutions involves many qualitative aspects. There are proxies: homicide rates for violence, perception surveys for safety and corruption, counts of unsentenced detainees or independent institutions for justice, etc. But these can be manipulated or not tell the whole story. Authoritarian countries often report low crime (sometimes true, sometimes due to fear of reporting) and high trust in institutions (state media narrative), which might look good on paper for SDG 16 but mask oppression. Even in open societies, perception-based metrics (e.g., surveys on trust or corruption) can fluctuate with events or biases. Data on issues like human rights abuses, illicit financial flows, or institutional independence are often contested and not officially reported by governments themselves (outside sources provide them). Many SDG 16 indicators are tier II or III (meaning data not regularly produced by countries). For example, tracking how many people have access to dispute resolution, or how well institutions are accountable, is tough. This allows countries to cherry-pick what they highlight – maybe the number of women in judiciary (an inclusion metric) goes up, which they'll report, but they won't report on political prisoners or something not required by SDG indicators. Without comprehensive and objective measures, an adversary can proclaim progress (like “we passed an anti-corruption law, indicator met”) while corruption still flourishes in practice. The complexity of governance also means even with data, attribution is tricky – a slight drop in violence could be due to many factors, not necessarily better institutions, but could be claimed as success.
- **Capture Resistance:** *Low.* By definition, Goal 16 is about reducing capture of institutions (making them accountable and inclusive). But those who currently capture institutions (corrupt officials, ruling parties, oligarchs) will resist. We see evidence: anti-corruption agencies being co-opted to target rivals, judiciaries packed with loyalists, media and civil society (key watchdogs) being intimidated or bought. In conflict zones, warlords or security forces may effectively capture any peace dividend. Internationally, efforts like anti-money laundering or tax transparency (to reduce illicit flows) get weakened by financial industry lobbies and haven countries that profit from the status quo. Building strong institutions is a slow, internal process; external imposition can't easily root out deep networks of patronage or vested interests. So capture attempts are rampant – in many countries, as reforms start, the entrenched elite either slow-roll them, water them down, or adapt to appear compliant while continuing old practices in new forms. For example, after cracking down on overt bribery, corruption might shift to more sophisticated embezzlement or nepotism, which is harder to pin down. The goal can be co-opted as well: a government might crack down on “criminals” vigorously (improving order on paper) but define criminals to include political dissidents – capturing the narrative of promoting peace to actually stifle opposition. In summary, Goal 16 goes against the grain of those benefitting from weak rule of law, so it likely faces internal sabotage by those actors.
- **Trust Degradation Risk:** *High (if seen as hypocrisy).* Governance goals are key to whether people trust their government and the social contract. If the public perceives that leaders have signed onto SDG 16 but continue to be corrupt, oppressive, or ineffective, it can breed cynicism and unrest. For instance, many countries have anti-corruption rhetoric (often more now due to international norms), but when scandals keep emerging and go unpunished, citizens lose trust not just in their government but in international initiatives that the government brandishes. On

the other hand, genuine improvements in safety or justice are quickly felt and boost trust. The risk here is that if by 2030 many countries still struggle with violence, injustice, and corruption – which is likely – people may question what the SDG process accomplished. It could feed narratives like “globalism doesn’t care about real democracy/human rights, it let dictatorships prosper as long as they tick other boxes.” Also, failing on Goal 16 can impede everything else, since conflict and corruption derail development – so a failure here undermines trust in achieving the whole Agenda. Conversely, there’s also a risk that aggressive pursuit of “peace” (like external interventions or heavy policing) could cause short-term grievances, but given this is about trust, the bigger concern is the former: the hypocrisy gap. Especially in conflict or post-conflict societies, broken promises of justice (like no reconciliation or accountability for atrocities despite global talk) can reignite tensions, the ultimate trust breakdown.

Survivability Score: 3/10. *This goal is highly political and often least prioritized because it challenges power structures directly. In an adversarial environment, progress is easily stymied or faked. While there are positive cases (some nations improving governance), overall trends on conflict, human rights, and corruption are not unequivocally positive in the 2015-2025 period; in some areas they worsened. The score is low because many actors have found ways to appear supportive of Goal 16 without real change, and global governance itself is strained (more conflicts erupted, democratic backsliding in many places). It’s a foundational goal – failure here cascades into other goals’ failure. So its low survivability reflects how susceptible it is to being hollowed out by the very forces it seeks to reform.*

Goal 17: Partnerships for the Goals

Original Formulation: “Strengthen the means of implementation and revitalize the global partnership for sustainable development.”

- **Symmetry Integrity: Moderate.** This goal is about global cooperation – finance, trade, capacity-building, systemic issues – ideally bridging the gap between developed and developing, public and private, etc. Symmetry would mean an even playing field: for instance, fair trade rules, adequate voice for poorer nations in financial institutions, mutual accountability in partnerships. In practice, partnerships often end up **asymmetrical**, with more powerful partners (donor countries, multinationals) setting terms and agendas, and weaker partners following. Adversaries could maintain those imbalances while invoking the language of partnership. For example, development aid might be tied to donor interests, or public-private partnerships might skew benefits to the private entity (privatizing profits, socializing risks). If a major economy wants to use Goal 17, it might highlight its aid contributions (even if they are below promised levels) and downplay any conditionalities that preserve asymmetry. Trade asymmetry persists (e.g., subsidies in rich countries while poor countries forced open), and while WTO reforms are talked about, powerful blocs may resist changes that reduce their advantage. On technology, symmetric partnership would share tech and data freely, but we often see knowledge monopolies (IP regimes favoring rich-world companies). Without equitable governance, partnerships can inadvertently become a means of extending influence or dependency – essentially rebranding the old donor-recipient model as “partnership.” That undermines the symmetrical, solidarity spirit that this goal aspires to.
- **Incentive Integrity: Low-Moderate.** The incentives for genuine global partnership are often trumped by national or corporate self-interest. Donor countries have incentives to count whatever they can as aid (to appear to meet targets) without necessarily providing new resources – e.g., including debt relief at face value, or refugee costs domestically, as part of aid figures. Many donors fell short of the 0.7% of GNI aid goal ²⁰, suggesting low incentive to

prioritize it. Instead, aid is often guided by strategic interests (geopolitical alignment, commercial ties). For international finance, wealthy nations control lending institutions and may not incentivize reforms that give emerging economies more say. Private sector “partnership” is incentivized when there's profit or PR to gain, but not for unprofitable gaps; companies might cherry-pick SDG areas that align with their business and neglect others, all while gaining reputational benefits. There's also incentive to push loans as “partnership” rather than grants, leading to debt issues (we see rising debt in developing countries, which could contradict the goal of sustainable finance). On systemic issues like tax cooperation (illicit flows), tax havens and the elites benefiting have zero incentive to change the system that advantages them. So while everyone rhetorically agrees partnership is good, practically each tries to minimize their own sacrifice: donors want to spend less, technology holders want to keep patents, trade powers want to keep their edge. The consequence is a lot of talk (forums, declarations) but less transformative action – an adversary could happily sign partnership MoUs which look great but commit them to little.

- **Measurement Hardness:** *Moderate.* Goal 17 has many targets but they are mostly about inputs and processes (finance amounts, number of agreements, capacity building, policy coherence indices). It's kind of a meta-goal, measured by e.g., volume of aid, FDI flows, debt sustainability measures, internet penetration (for technology), etc. Those are quantifiable to an extent. But measuring the *quality* of partnerships or whether they are “revitalized” is fuzzy. A donor can meet a numeric target by some accounting creativity; whether that money truly helps achieve SDGs is harder to trace. Similarly, counting South-South cooperation initiatives or multi-stakeholder partnerships by number doesn't tell you if they're effective or one-off conferences. Many metrics rely on self-reporting by governments (like how integrated SDGs are in policy – which any government can claim “we have an SDG strategy” regardless of actual integration). It's easy for adversaries to inflate these: announce big initiatives that span multiple partners, which ticks the partnership box, but might be dormant later. Without qualitative evaluation, one could boast increased partnerships even if outcomes are lacking. Also, some crucial aspects like policy coherence or systemic issues (like tax justice) are not captured well by simple indicators, so those can slip by without showing up as a failure in Goal 17 metrics. Thus, a country could conceivably get good marks on Goal 17 (for hosting partnerships, contributing some aid, improving data collection, etc.) while still engaging in practices counter to the spirit (like protectionist trade policies or insufficient support). The measurement focus on means rather than ends makes it easier to claim success here than on other goals that have concrete outcomes.

- **Capture Resistance:** *Low-Moderate.* The concept of partnership is broad; it can be captured by dominant players to push their agendas under a cooperative guise. For instance, a global partnership on infrastructure might largely serve the interests of the corporations building projects and the donors financing them (exporting their construction services, etc.), with less regard to local needs – but it's branded as mutual development. There's risk of multistakeholder partnerships where civil society or local voices are token participants compared to government and business interests – capture by imbalance of power within the partnership. Internationally, “global partnership” might be interpreted by major powers as them setting rules that others should join (some argue current global institutions themselves reflect such capture historically). Also, with the rise of rival geopolitical blocks, each might try to redefine partnership: e.g., one Belt-and-Road type partnerships vs Western-led alliances – each potentially using SDG rhetoric. That fracturing could capture Goal 17 into fragmented spheres rather than a truly global approach. On data (one target is about data monitoring), big tech or donor-driven approaches could capture the data agenda, focusing on what they think is important rather than country-owned priorities. However, among SDGs, partnership is one where if captured, it often just means not much happens beyond status quo, since it relies on voluntary cooperation. So capture

here is more like co-optation – using the language to advance existing power structures – rather than overt sabotage. Still, that co-optation can block the transformative potential (like global tax reforms or equitable trade deals).

- **Trust Degradation Risk: Moderate.** This goal underpins the credibility of the entire SDG enterprise: if global partnership fails, none of the others can fully succeed. Trust between countries – especially developing and developed – hinges on honoring commitments (finance, technology, fair policies). So far, partial failures (like unmet aid promises, vaccine inequity recently, etc.) have eroded trust. Developing countries often voice frustration at big talks but slow action from rich nations, which can hamper cooperation on other fronts (e.g., climate negotiations where finance trust gap is a big sticking point). If by 2030 the promise of revitalized partnership rings hollow – say, if aid remains low, debt crises mount, and trade remains skewed – the global South’s trust in multilateralism will erode further, potentially leading to a more fractured international system. Domestically, if people see that promised international support or cooperation isn’t materializing (like financing for projects or access to medicines), they may also doubt global initiatives. Conversely, too much emphasis on partnership with private sector without results can cause public skepticism (“corporations were at the table but nothing improved for us”). However, since this goal is somewhat abstract for the general public, trust effects might manifest indirectly via the success or failure of other goals. A notable risk is that populists or nationalists can exploit any shortfall in global partnership: “See, the UN and global elites don’t help us; we must go it alone.” Such sentiments are already visible. Therefore, failing to deliver tangible solidarity might strengthen isolationist narratives. On the flip side, some effective partnerships (like swift collaboration in a crisis) could boost trust temporarily. Overall, moderate risk: significant to policy insiders and international relations, but less immediately felt by a citizen compared to say lack of water or jobs.

Survivability Score: 5/10. *As a means-goal, Goal 17 has had a mixed performance. Some areas improved (e.g., more data availability, some technology cooperation), others lagged (financial commitments, systemic trade/tax reforms). It’s somewhat insulated from direct adversarial attack because no one openly opposes “partnership,” but that also means it’s prone to benign neglect or lip service. We score it around mid: the framework of partnerships will survive (people will keep talking about it), but the efficacy is limited. The goal can absorb a lot of lackluster outcomes while still being officially deemed ongoing. Its survival is less dramatic (it won’t “fail” visibly like not ending hunger would) but rather fade into a bland call with modest impact if adversarial/selfish behaviors persist. Realistically, by 2026 we see major strains in multilateralism, but also efforts to reinvigorate it (post-pandemic). So it hangs in the balance – not fully undermined, but far from fully realized.*

Survivability Scores Summary

The following table summarizes the **survivability score** for each SDG based on our adversarial analysis. Scores are on a **1–10 scale** (10 = extremely robust/resilient to adversarial challenges; 1 = extremely vulnerable/likely to be subverted). We also list key vulnerabilities identified for quick reference:

SDG	Goal (Abbreviated)	Survivability Score	Key Adversarial Vulnerabilities
1	No Poverty	4/10 (Low)	Data manipulation; neglect of ultra-poor; superficial progress claims; non-binding commitments.

SDG	Goal (Abbreviated)	Survivability Score	Key Adversarial Vulnerabilities
2	Zero Hunger	3/10 (Low)	Tragedy-of-commons in food; yield-over-nutrition focus; agribusiness capture; conflict & climate risks.
3	Good Health & Well-being	5/10 (Medium)	Focus on metrics > quality; pharma/donor agenda capture; inequality in access; pandemic preparedness gaps.
4	Quality Education	4/10 (Low)	Quantity > quality (learning crisis) ¹⁵ ¹⁶ ; unequal access; curriculum politicization; data gaps on outcomes.
5	Gender Equality	5/10 (Medium)	Tokenism; cultural resistance; selective empowerment (urban elites vs rural women); weak enforcement of rights.
6	Clean Water & Sanitation	6/10 (Med-High)	"Last mile" neglect ⁷ ; maintenance failures; unequal service quality; potential privatization capture.
7	Affordable & Clean Energy	5/10 (Medium)	Greenwashing (gas as "clean"); focus on access over affordability; fossil lobby influence; uneven transition burdens.
8	Decent Work & Economic Growth	3/10 (Low)	GDP-centric incentives ¹⁷ ; ignoring job quality; corporate capture; conflict with environmental limits.
9	Industry, Innovation & Infrastructure	4/10 (Low)	Crony projects; urban-rural divides; focus on big builds > inclusive infrastructure; corruption in contracts.
10	Reduced Inequalities	2/10 (Very Low)	Elite resistance/capture; lip service vs redistributive action; global inequities in aid/trade persist ² .
11	Sustainable Cities & Communities	5/10 (Medium)	Gentrification; slum neglect; developer capture; superficial "smart city" solutions over basic services.
12	Responsible Consumption & Production	3/10 (Low)	Corporate greenwashing ²¹ ; consumerism intact; weak metrics; externalization of waste to poorer areas.
13	Climate Action	4/10 (Low)	Free-riding on emissions cuts; fossil lobby delay; unmet finance pledges; slow mitigation despite urgency ²⁰ .
14	Life Below Water	4/10 (Low)	Overfishing incentives; weak high-seas governance; pollution outsourcing; marine resource exploitation by big players.

SDG	Goal (Abbreviated)	Survivability Score	Key Adversarial Vulnerabilities
15	Life on Land	4/10 (Low)	Deforestation profit motives; wildlife trafficking; land use conflicts; conservation without community buy-in.
16	Peace, Justice & Institutions	3/10 (Low)	Authoritarian backlash; corruption entrenched; façade reforms; conflict hotspots remain unstable.
17	Partnerships for Goals	5/10 (Medium)	Donor fatigue; unequal partnerships; lots of rhetoric, limited new resources; geopolitical tensions.

Interpretation: Scores in the **1–4 range indicate goals at high risk** of failure or co-option under adversarial conditions, needing serious overhaul. Scores around **5–6 indicate moderate resilience** – some progress likely, but still significant vulnerabilities. No goal scored above 6/10, reflecting the across-the-board challenges in a non-cooperative scenario. In particular, **SDG 10 (Inequality)** emerged as a critical weak link (scoring 2/10), often called a linchpin for stability – its poor performance undermines many others. Conversely, **SDG 6 (Water)** appears relatively more tractable (6/10) due to clear outcomes and localizability, though not without issues ⁷.

These scores underscore that **business-as-usual will leave most goals unmet and easily undermined**. A reimagined approach is required to boost each goal's survivability – which we present next through reconstructed alternatives.

Reconstructed SDGs: Adversarial-Resilient Alternatives

In this section, we present **reconstructed versions of all 17 SDGs** designed to withstand the adversarial challenges identified. Each rephrased goal is crafted to meet the six constraints outlined (Symmetry-Preserving, Outcome-Bound, Adversarially Auditable, Capture-Minimizing, Failure-Visible, Trust-Restorative). These formulations aim to preserve the spirit of the original SDGs while addressing their implementation gaps and exploitability. Accompanying each goal, we note how the new framing and associated strategies fulfill the constraints:

1. **No Poverty (Reconstructed):** *“Achieve zero extreme poverty and reduce relative poverty by 50%, through inclusive economic security measures verifiable at household level in every community.”*
Why & How: This sets a **clear outcome** (zero people below an extreme poverty line, and halved relative poverty) rather than the broad “end poverty in all forms.” It demands disaggregated verification (by community), ensuring **failure visibility** if any pocket lags. By including *relative poverty*, it **preserves symmetry**, addressing inequality within societies, not just an average lift. Implementation would include independent poverty audits (e.g., statistically validated surveys with civil society oversight) – making it **adversarially auditable** (harder for governments to fake numbers ²² ²³). To **minimize capture**, the goal calls for direct economic security measures (such as cash transfers, universal basic needs services) that reach the poorest transparently, reducing the chance that only elites benefit. These mechanisms, coupled with public reporting on progress in each locality, are **trust-restorative** – people can see and confirm poverty reduction in their own community, or flag issues if the promised support isn't arriving.

2. **Zero Hunger (Reconstructed):** *“End hunger and malnutrition by ensuring universal access to sufficient and nutritious food, with priority to the most food-insecure populations and agro-ecological methods that are monitored for sustainability.”*

Why & How: This wording emphasizes not just ending caloric hunger but also *malnutrition*, making the **outcome-bound** target about adequate nutrition for all. It explicitly prioritizes the most food-insecure (e.g., children, conflict zones) – a **symmetry-preserving** approach so no group is left behind for efficiency’s sake ⁷. It also ties food security to sustainable agriculture (“agro-ecological methods”), addressing the tension between yield and environmental health; these methods can be audited (e.g., tracking soil health, biodiversity on farms) to ensure **adversarial auditability** of sustainability claims. By monitoring outputs like local nutritional status (stunting, anemia rates) and farming impact, any **failure is visible** if, say, yields rise but malnutrition remains or if production gains come at ecological cost. **Capture-minimization:** the focus on agro-ecology and local food security measures (like community food reserves or climate-resilient crops) diversifies away from reliance on a few agribusiness giants, and international oversight (via FAO or others) of hunger hotspots prevents political cover-ups of famine. All these build **trust**, since the public sees concrete reductions in hunger and knows that both quantity and quality of food are addressed without exploiting the environment for short-term gains.

3. **Good Health & Well-Being (Reconstructed):** *“Achieve universal health coverage with equitable access to essential services and a 50% reduction in preventable deaths, verified by independent health outcome audits.”*

Why & How: This reframing makes the goal **outcome-bound** by specifying measurable targets: universal health coverage (UHC) and halving preventable mortality (maternal, infant, major diseases). UHC implies everyone is covered by healthcare – a clear, countable outcome (either via insurance or basic services). The 50% reduction gives a concrete benchmark to hit for key health outcomes, making failure harder to hide if rates don’t decline. **Symmetry-preserving:** emphasis on equitable access ensures improving averages isn’t enough – disparities (urban/rural, rich/poor) must be closed; independent audits would check service coverage in remote or marginalized areas (no last-mile neglect). **Adversarially auditable:** independent health outcome audits mean that data on deaths, disease incidence, etc., would be cross-verified by perhaps academic institutions or international bodies, not just governments, reducing the chance of cooking the books ²⁴ ¹³. Also, disaggregated data (by region and group) must be published. **Capture-minimizing:** focusing on essential services (e.g., primary care, vaccinations) steers resources to public health basics rather than allowing capture by specialty or high-tech hospital interests that might serve only a few. Involving communities in health boards and monitoring patient feedback loops further ensures institutions serve the public, not bureaucratic or corporate interests – fostering **trust**. People will tangibly experience improved health services and see independent confirmation of better health outcomes, which reinforces belief in the system.

4. **Quality Education (Reconstructed):** *“Achieve universal functional literacy and numeracy, and ensure all youth attain recognized secondary-level skills, with progress measured by independent learning assessments.”*

Why & How: Instead of vaguer “quality education,” this zeroes in on **outcomes:** functional literacy and numeracy for all (basic foundation), and secondary-level skills (academic or vocational) for youth. These are concrete and testable – a student either reaches a proficiency or not – making the goal **outcome-bound** and **auditable** through standardized assessments (preferably run or validated by neutral observers, not solely by the education ministry). By focusing on *learning* (not just schooling), it counters the incentive to just increase enrollment. **Symmetry-preserving:** universal means everyone, so the system must target marginalized

groups (girls, disabled, minorities) to truly get 100% literacy – not just national average. If any subgroup lags, goal is not met, which pressures governments to address inequities (or it becomes **visible failure** when disaggregated literacy rates are published ¹⁵ ¹⁶). **Capture-minimizing**: specifying “recognized skills” at secondary means curricula should meet standards set by broad consensus (including employers, communities) rather than any one ideological agenda – reduces scope for politicizing content. Also, independent assessments curtail any temptation for authorities to lower standards to claim success. The **trust** comes as parents and students see genuine improvements in competencies (not just paper qualifications). Internationally comparable assessments (like a global PISA for SDGs) could verify progress and shame cheaters. Knowing that every child can actually read, do math, and has some marketable skill by end of schooling would be a powerful confidence boost in the education system’s integrity.

5. Gender Equality (Reconstructed): *“Attain equal representation and empowerment of women in social, economic, and political life, and eliminate gender-based violence, monitored via gender-disaggregated outcomes and independent rights observance.”*

Why & How: This formulation is **outcome-bound** by calling for equal representation (e.g., 50/50 in political bodies, management, etc. – symmetry in participation) and empowerment in measurable spheres like workforce participation, property ownership, education levels, etc. It also sets the elimination of gender-based violence (GBV) as a clear target (outcome of safety and dignity). These can be tracked: representation by numbers, GBV by prevalence surveys.

Symmetry-preserving: explicitly about equality of outcomes between genders, not just improvement for women in isolation; ensures issues where men lag (if any, e.g., boys’ education in some areas) can be addressed too, heading off narratives of one-sidedness. **Adversarially auditable:** requiring gender-disaggregated outcomes for all relevant indicators (income, health, education, etc.) means any disparity stands out in reports – a government can’t hide marginalization behind averages. Also, independent rights observance implies something like periodic reviews by expert bodies or civil society (e.g., CEDAW committee or similar) to verify on-the-ground status of women’s rights, adding an audit layer beyond government self-reporting.

Capture-minimizing: This framing is broad but with specific yardsticks (representation, GBV, economic equality), so a government or organization can’t just empower a tiny elite of women and claim victory – because data would show if most women still lack rights or safety. It also can’t cherry-pick one aspect (like corporate board seats) while ignoring, say, domestic violence, because GBV elimination is part of the goal and must be addressed. Involving women’s organizations in monitoring and implementation helps guard against token measures and builds **trust**. Women and the public see real changes: more women in decision-making, fewer incidences of violence – outcomes that restore faith that this is not mere “moral signaling” but actual social transformation.

6. Clean Water & Sanitation (Reconstructed): *“Guarantee universal access to safe drinking water and sanitation services on premises, with priority maintenance for vulnerable communities and transparent water quality monitoring.”*

Why & How: This retains a clear **outcome** – everyone has water and sanitation at home (not just within X distance, but on premises, reflecting true accessibility). It emphasizes *safe* drinking water (quality aspect) and adds that services must be maintained (not just built once).

Symmetry-preserving: “priority maintenance for vulnerable communities” means resources first go to areas that historically had gaps (rural villages, informal settlements) to ensure they catch up, and that they aren’t left with broken systems while richer areas get continuous upgrades ⁷ . So it keeps focus on equal service levels. **Adversarially auditable:** transparent water quality monitoring means frequent publication of water safety test results and functionality data (e.g., how many systems are working) – possibly crowdsourced or via third-party audits. If a local

water system fails, that data should be visible quickly, creating pressure to fix it (**failure-visible** in real time). **Capture-minimizing**: by framing it as a guaranteed public service, it implies governments and communities retain control and accountability for water – private entities can play a role, but under regulation to meet the guarantee. This reduces risk of a private concessionaire cherry-picking profitable areas or raising tariffs too high; if they do, it violates the “guarantee” and triggers government action. Publishing performance metrics (continuity, quality, complaints) holds providers accountable regardless of who they are. People knowing that the government actively tests their water and fixes issues builds **trust** that their health is safeguarded and no community is second-class. Essentially, this sets a high bar (like human-right-to-water in practice) and demands evidence to prove it’s met.

7. Affordable & Clean Energy (Reconstructed): *“Secure universal access to reliable and affordable energy with at least 80% coming from low-carbon sources by 2030, audited via transparent emission and pricing data.”*

Why & How: This couples energy access with a clean energy transition target (80% low-carbon by 2030 – ambitious but forces action). The outcome is twofold but measurable: (a) access for all (could be measured by electrification rate and clean cooking access), and (b) energy mix decarbonization to 80%. Those give **outcome-bound** clarity rather than just “promote renewables.” **Symmetry-preserving**: stresses affordable for all, meaning policies must ensure the poor can pay (subsidies, tiered tariffs), not just connection. Also, universal means urban and rural, rich and poor alike get reliable power – no region left dark. By including a high low-carbon percentage, it signals rich countries especially must cut fossil use drastically (they can achieve that easier) – addressing global symmetry where developed nations carry more responsibility to decarbonize quick, while also enabling developing ones to leapfrog to clean tech (with support). **Adversarially auditable**: transparent emission and pricing data – i.e., publish the CO₂ per kWh of generation, fuel mix, and monitor energy prices relative to income. This data being open lets watchdogs and the public verify that the transition is real (no creative accounting of “clean”) and that energy remains affordable. If emissions from energy aren’t dropping or if prices spike beyond reach, it’s immediately evident and attributable, preventing false claims. **Capture-minimizing**: a clear numeric clean target limits the fossil fuel lobby’s ability to delay – governments either diversify to renewables or miss the target publicly. Having transparency in pricing also discourages collusion or undue profit-taking by utilities, since the public would see if costs are unjustified. It fosters **trust** by promising a modern, green energy system accessible to everyone, and by verifying progress through data (for instance, citizens can see local air pollution drop and their bills stable as renewables scale up, reinforcing that the transition is working for them).

8. Decent Work & Economic Growth (Reconstructed): *“Ensure sustainable economic progress by raising median incomes by at least 40% and securing full employment under fair labor conditions, without exceeding per-capita environmental limits.”*

Why & How: This reframing pivots from GDP to median income growth and full employment with labor standards. “Raising median incomes by 40%” focuses on the typical worker’s benefit (if GDP grows but median doesn’t, goal fails), thus addressing **inclusion** and being **outcome-bound** in terms of people’s actual earnings. “Full employment” (meaning ~everyone who wants a job can get one) under fair conditions (minimum wage, safety, rights) directly addresses decency of work. The environmental caveat “without exceeding per-capita environmental limits” injects sustainability – e.g., keep carbon or material footprint per person within a target. That aligns growth with environmental responsibility quantitatively, tackling the prior contradiction ¹⁷. **Symmetry-preserving**: median focus and full employment inherently aim to lift the lower half of society and include marginalized workers, not just enrich the top or accept jobless growth. Also, linking to environmental limits means rich countries must adjust growth patterns to be greener

(so global symmetry in resource use improves) – no more offloading pollution. **Adversarially auditable:** median incomes and employment rates are solid metrics, less easy to fudge than GDP (median income comes from surveys but can be verified independently by household data). Environmental footprint per capita can be monitored (carbon per capita is straightforward, others like material footprint approximated), ensuring **failure visible** if progress comes by unsustainable means. **Capture-minimizing:** by emphasizing labor conditions and median outcomes, policies can't just cater to big corporations; in fact, they will require empowering labor (e.g., stronger collective bargaining, fair wage laws) which counters corporate capture of all productivity gains. Environmental limits clause means industries can't capture policy to pollute freely – progress measured isn't just economic, but coupled with environmental health. These help build **trust:** workers see their livelihoods improve and are not asked to choose between jobs and environment – both are delivered. It reframes “growth” not as an end but as a means to shared prosperity within planetary boundaries, which is a narrative citizens can trust because it binds leaders to multiple accountability points (social and ecological) ¹⁹ .

9. Industry, Innovation & Infrastructure (Reconstructed): *“Build resilient infrastructure and expand sustainable industrial output by 50%, prioritizing underserved regions and green technologies, with open monitoring of project quality and impact.”*

Why & How: This sets quantifiable expansion (“industrial output +50%” – could be manufacturing value-added or similar) but crucially attaches conditions: must be sustainable (e.g., low-carbon, resource-efficient) and resilient infrastructure. “Prioritizing underserved regions” ensures **symmetry** – investments go first to places lacking infrastructure (lagging regions, rural areas) so benefits balance out spatial inequalities, rather than just boosting existing industrial hubs. **Outcome-bound:** 50% industrial output growth is concrete, and resilient infrastructure built can be measured in coverage (e.g., % roads paved, % population with internet) – focusing on outcomes (access provided, capacity built) not just money spent. **Adversarially auditable:** open monitoring of project quality and impact means data on infrastructure projects (cost, contractors, completion time, environmental compliance, utilization rates) are published for scrutiny, reducing corruption and “bridges to nowhere.” Similarly, innovation/industry impacts (like jobs created, emissions from new industries) should be tracked openly. This transparency makes **failures visible** (e.g., if a factory pollutes a river or a highway collapses, it can't be swept under rug) and lets citizens flag if infrastructure isn't meeting needs. **Capture-minimizing:** emphasis on green tech and underserved areas inherently diverts from just profit-driven projects in wealthy enclaves. Publishing procurement and impact data deters corporate capture via shady deals, as watchdogs can catch anomalies. Multi-stakeholder oversight committees (including local community reps and engineers) could be mandated for big projects to ensure resilience and inclusion criteria are met. All this improves **trust** – communities see new, useful infrastructure reaching them (not just elites), built to good standards (no shoddy construction due to corruption), and industries are adding jobs without harming the environment grossly. When people physically experience better connectivity (roads, internet, power) and local economic activity that respects the environment, they gain confidence in “industry and infrastructure” as positive forces, not exploitative ones.

10. Reduced Inequalities (Reconstructed): *“Reduce income and wealth inequality by elevating the bottom 40%’s share of national income by 25%, through policies of redistribution, inclusion, and anti-discrimination verified by independent equity audits.”*

Why & How: This directly targets a measurable **outcome:** the income share of the bottom 40% (a common inequality indicator) should increase by 25% relative to baseline – a proxy for a more equitable distribution ²⁰ . One could also include Gini reduction or wealth share improvements, but bottom 40% share is tangible and speaks to uplifting the lower segment. Achieving this likely requires redistribution (tax/transfer), inclusive growth (jobs for the poor), and tackling

discrimination that keeps certain groups poorer – which the goal text explicitly calls out as methods, signaling governments must implement those. **Symmetry-preserving:** It's basically the symmetry goal – bringing those at bottom closer to those at top. It also inherently involves ensuring all groups (by gender, ethnicity, region) improve because bottom 40% usually overlaps with marginalized groups. The mention of anti-discrimination highlights that aspect of symmetry across social divides. **Adversarially auditable:** independent equity audits would be regular reviews by perhaps national auditors or international observers that look at policy impacts on inequality – e.g., tax incidence analyses, benefit incidence for government spending, compliance with inclusive practices in hiring. They would publish findings on whether inequality is truly dropping and if policies are benefiting the intended groups. If a government enacts a flat tax cut that helps the rich, the audit would flag that contradiction. These audits make **failure visible** yearly rather than waiting to see final outcomes. **Capture-minimizing:** explicit focus on redistribution and audits reduces wealthy capture – e.g., tax loopholes or regressive subsidies would be scrutinized. Also, involving civil society (NGOs focused on equality, perhaps an independent ombudsperson) in the audit or policy design can counterbalance elite lobbying. **Trust-restorative:** inequality is a root of social distrust; if people observe a genuine narrowing of gaps – rising wages for the poorest, reduced extreme wealth concentration – they will feel the system is fairer. That renews trust in institutions because it shows they're acting in the common interest, not just for elites. The independent audits add credibility because progress reports come from a trusted referee, not just government boasting. Achieving visible reductions in inequality would powerfully signal that society is becoming more just, addressing one of the biggest cynicism drivers.

11. **Sustainable Cities & Communities (Reconstructed):** *“Achieve inclusive urban development by upgrading 100% of slums and ensuring 70% of urban trips are by affordable public transport or walking/cycling by 2030, with local community oversight of planning.”*

Why & How: This gives two concrete targets: (a) upgrade all slums (so no one lives in substandard housing) – that's **outcome-bound** and directly addresses inclusion/symmetry, and (b) shift urban mobility to 70% sustainable modes (public transit or non-motorized) – which addresses sustainability and livability. These are bold but measurable: slum upgrading can be tracked by number of households in slums over time (target zero un-upgraded slums), and transport mode share can be measured via surveys or transit usage stats. **Symmetry-preserving:** focus on slums explicitly prioritizes the poorest communities in cities. Also, affordable public transport aims to provide mobility for all, not just car owners, knitting the city together. If 70% trips are green modes, it implies even middle class uses public transit – indicating it's high quality (reducing class-based disparities in transport). **Adversarially auditable:** local community oversight means residents participate in planning and monitoring upgrades – they'll call out if a project fails (e.g., a housing project built on paper but not delivered, or poor construction). Also, public transport improvement progress (new lines, fare levels, ridership) can be transparently reported city-wide, making any neglect of segments visible (e.g., if only city center gets transit and peripheries don't, data will show low transit share there). **Capture-minimizing:** involving communities in decision-making helps counter capture by developers – plans have to be justified to people, not just behind closed doors. Requiring full slum upgrade means the city can't just focus on flashy business districts (developers prefer that); they must invest in the least profitable areas (often city-owned upgrades or sites where private developers wouldn't go without incentives). The transport target reduces car-centric development push by auto and oil industries, and encourages investment in public goods. **Trust:** when former slum residents see their neighborhood improved (with services, secure tenure, decent housing) and urban dwellers see reliable, cheap transport for everyone, it greatly increases trust in local government. It shows the city values all citizens, not only the rich. Additionally, participatory planning means people trust that their voices count in urban

development. Achieving these would make cities visibly more inclusive and sustainable, proving that global goals delivered concrete benefits in daily life.

12. **Responsible Consumption & Production (Reconstructed):** *“Cut per-capita waste and pollution by 50% and ensure all major supply chains are transparent and circular by 2030, enforced through independent environmental and social audits.”*

Why & How: This tackles both the consumption side (waste/pollution per person halved – e.g., food waste, plastic waste, CO₂ emissions, etc. aggregated or separate targets) and the production side (supply chains transparent – so one can see impacts – and circular – meaning high recycling/reuse, minimal virgin resource use). Those are **outcome-bound**: numeric reduction in waste/pollution and a qualitative state (circularity) that can be measured by metrics like % recycled materials, product life extension, etc. Transparency and circularity for all major supply chains (like food, clothing, electronics, etc.) means companies must disclose sourcing and end-of-life handling, enabling **audits** of their practices. **Symmetry-preserving**: reducing per-capita waste/pollution implies high-consuming populations must curb excess strongly – leading to more equitable resource use globally. It also means less dumping of waste on poorer regions (can't meet waste reduction if just exporting it). Transparent supply chains prevent externalizing harms to weaker regions or workers – consumers in rich countries can no longer be ignorant of conditions in poor producer countries, helping fix global asymmetries of knowledge and power. **Adversarially auditable**: independent audits (could be by NGOs or certified bodies) of environmental and social aspects ensures companies aren't marking their own homework. Because supply chain transparency is mandated, watchdogs can trace if a product was made with deforestation or forced labor. Circular economy claims (like “our packaging is 100% recyclable”) can be verified by actual recycling rates – making **failure visible** if recycling infrastructure or corporate take-back doesn't actually happen. **Capture-minimizing**: mandatory transparency and third-party audits take away the option of corporate self-reporting only the good news. It also levels the field – if all must follow, no company gains a short-term advantage by cheating without exposure. It turns business lobbies towards innovating in circularity rather than fighting regulation, since the targets are firm. Government performance is also subject to audits (e.g., is waste management actually improved or just hidden?). While this goal is ambitious, achieving even most of it would **restore trust**: consumers trust that products are not harming people/planet unseen, citizens trust that big polluters are being held accountable, and communities trust that waste won't flood their land because overall waste is down. It shows a serious shift from exploitative production to stewardship – a moral and practical reset that can inspire confidence in global cooperation.

13. **Climate Action (Reconstructed):** *“Limit global warming to 1.5°C by achieving a 50% reduction in greenhouse gas emissions by 2030 and fully funding adaptation for vulnerable regions, with progress verified by real-time emissions monitoring and climate risk audits.”*

Why & How: This states the Paris Agreement's toughest target (1.5°C) as the goal and quantifies the immediate requirement: 50% cut globally by 2030 (which can be downscaled per country in fairness). It also adds the adaptation funding commitment – often neglected – to protect vulnerable regions, making the goal **symmetry-preserving** (those who pollute fund those who suffer more). **Outcome-bound**: yes, 50% emissions cut is very concrete and time-bound; hitting 1.5°C is the ultimate outcome (though cannot be directly measured until mid-century, the emissions cuts are the proxy outcome needed now). **Adversarially auditable**: real-time emissions monitoring (via satellites, etc.) means we don't rely only on self-reported inventories years later – we track large emissions sources (power plants, deforestation) as they happen ¹⁹. This makes any lack of action **visible immediately** (e.g., if a country keeps high coal output, satellites and other data will show CO₂ not dropping). Climate risk audits would assess how well countries/vulnerable communities are being protected (e.g., early warning systems, resilient

infrastructure) – likely by independent agencies or UN reviews. If a cyclone hits and devastation is huge, audit might reveal adaptation gaps and who failed to finance them. **Capture-minimizing:** a single stark target for emissions leaves little room for fossil fuel interests to wiggle – they must either transform or fade; incrementalism or deceptive “plans” would be outed by the hard numbers. Internationally, making adaptation finance an explicit part calls out rich nations to fulfill promises – audits can check if money flowed and projects implemented. By closely monitoring, any attempt to use offsets or accounting tricks gets caught if atmospheric levels or sectoral trends don’t match claims. **Trust-restorative:** actually doing what’s needed on climate would be a huge trust win with the global youth and vulnerable nations who feel betrayed currently ⁸. Real-time transparency also builds trust – the public can see emissions falling (or call out if not). And delivering funds to help poor regions adapt shows solidarity in action, potentially easing North-South tensions. If by 2030 people witness noticeable climate action (e.g., clean energy everywhere, declining emissions, communities safer from floods), it will revive faith that governments can tackle even the toughest problems collaboratively, rather than just talk and posture.

14. **Life Below Water (Reconstructed):** *“Restore ocean health by ending overfishing (100% fish stocks at sustainable levels) and protecting 30% of marine habitats by 2030, enforced with satellite surveillance and international inspections.”*

Why & How: This gives two clear outcomes: no overfishing (every fish stock either recovered or fished within scientific quotas) and 30% of oceans protected (the “30x30” target agreed by biodiversity conferences). These are **outcome-bound** (specific percentages). **Symmetry-preserving:** applying to all fish stocks and having international enforcement means big fishing nations can’t exempt themselves – a fair rule for all. Small countries’ waters get protection, and they get assistance to enforce, balancing against large illegal fleets. Marine protected areas (MPAs) include high seas, not just territorial waters, sharing responsibility. **Adversarially auditable:** satellite surveillance (already used to track ships) plus inspections (e.g., port state control for catches, and potentially an international patrol for high seas) mean that illegal fishing or reserve violations can be detected and acted on quickly, rather than relying on self-report. Quota adherence can be checked against independent stock assessments and catch data. If a country overfishes or doesn’t police its boats, that will show in surveillance data or fish population trends – making **failures visible** (e.g., fish biomass indices not improving triggers an alert). **Capture-minimizing:** by having international oversight, we reduce risk of industry pressure within a single country quietly raising quotas. If quotas are set via transparent scientific bodies and catches monitored, industry must comply or face sanctions (market access denial, fines, etc.). The 30% protected areas, if globally monitored, prevents exploitation in at least those zones; to avoid capture in selection (only making parks in useless areas), criteria ensure important biodiversity areas are included. **Trust-restorative:** when communities see fish returning (many coastal folk intimately notice when fisheries rebound) and see that foreign vessels or big trawlers are kept out of their waters, they regain trust in governance. Global trust is bolstered if countries actually hold each other accountable at sea – often a lawless domain. The ocean’s recovery (more fish, cleaner waters, stabilized coral) is something tangible but requires collective discipline; achieving it would be a huge proof that multilateral action works. Public campaigns and citizen science (like divers monitoring reefs) can complement official monitoring, further boosting credibility and involvement.

15. **Life on Land (Reconstructed):** *“Halt biodiversity loss by achieving zero human-driven extinctions and a net gain of 10% in native forest area by 2030, secured by community-based conservation and transparent land-use governance.”*

Why & How: Two outcomes: zero extinctions (none of the known species go extinct due to human causes) – which is bold but gives a clear marker of success/failure, and a net 10%

increase in native forests (so not plantations, but actual forests regrowing) – addressing climate and habitat. These are **outcome-bound** and easy to understand. **Symmetry-preserving:** community-based conservation centers indigenous and local people in stewardship, recognizing their rights and knowledge, which corrects asymmetries where top-down approaches dominated ¹⁸. Transparent land governance (e.g., publicly accessible maps of concessions, land tenure registries) prevents companies or elites from secretly grabbing land or deforesting. It empowers citizens in all countries to know what's happening and hold perpetrators accountable. **Adversarially auditable:** extinction and forest cover can be monitored (e.g., IUCN Red List status updates, satellite forest maps). If deforestation happens or species populations collapse, it's visible data. Transparent land-use means things like deforestation alerts, public deforestation tracking online (already tools like Global Forest Watch) – so any deviation from the net gain trajectory is caught immediately (country X lost Y hectares – needs correction). Independent ecological audits would verify if “net gain” isn't being faked by planting monocultures. **Capture-minimizing:** land-use transparency undermines corrupt deals; if a logging concession is given in a protected area, everyone will know. Community conservation means those living on the land have veto power or co-management, reducing outside capture by big agribusiness or logging interests. Also, a zero extinctions ethic forces even powerful industries to avoid the most critical habitats or mitigate heavily, since letting a species die out would be a glaring failure. **Trust-restorative:** indigenous and local communities start to trust that global processes respect them (as they directly benefit from rights and support for conservation efforts). The public, especially youth, see real forest recovery and no more heartbreaking extinctions of charismatic or any species, which builds hope. Knowing that the tiger, pangolin, or orchid was saved because of concerted effort validates global action. It shows environmental and human interests aligned (no fortress conservation harm since communities are in charge). All together, if people witness deforestation reversed and wildlife returning, it combats eco-doomism and proves that sustainability is achievable, increasing trust in institutions tasked with safeguarding nature.

16. **Peace, Justice & Strong Institutions (Reconstructed):** *“Build just societies by reducing violent deaths by 75%, ensuring 100% birth registration and legal identity, and eradicating corruption through open governance and independent judiciary oversight.”*

Why & How: This picks specific measurable proxies for peace (violent deaths down – covers war and crime) and justice/inclusion (everyone has legal identity – a gateway to rights; also indicates state presence, no one off the grid). Eradicating corruption is qualitative, but we can measure through indices or number of high-level convictions etc. The means are stated: open governance (transparency, e.g., open budgets, open contracts) and independent judicial oversight (which implies judges and auditors can act without political interference). **Outcome-bound:** 75% reduction in violent deaths (e.g., from X per 100k to Y) is clear. Birth registration/legal ID to 100% is clear (no one stateless or uncounted). These put focus on tangible improvements in safety and inclusion rather than broad “promote peace.” **Symmetry-preserving:** legal identity for all ensures even marginalized groups are recognized by the state. Reducing violence broadly benefits especially those in conflict zones or high-crime poor neighborhoods (since often the wealthy can shield themselves, this focuses protection on all lives equally – if overall deaths plummet, likely the most vulnerable saw the biggest improvement). Anti-corruption and open governance level the playing field between elites and citizens – information symmetry and equal application of law. **Adversarially auditable:** violent death rates are tracked by health and police data (and even independent counts in war by NGOs) – hard to fully hide a war or crime wave. If a government tries to fudge crime stats, independent surveys or media can often countercheck. Requiring *independent judiciary oversight* means institutions like a supreme court or anti-corruption commission must publish findings, adding to accountability (they might release an annual report on corruption cases or on government abuse of power – making **failures visible** even if executive tries to hush). Open governance means data on budgets, spending, legislation,

are out in the open – any capture or injustice (e.g., biased public spending, missing funds) can be spotted by civil society. **Capture-minimizing:** these measures – transparency, strong judiciary – are classic antidotes to capture. Harder for a small group to monopolize power when judiciary can check them and finances are transparent. Also, universal birth registration prevents neglect of minorities (sometimes lack of IDs is used to marginalize). Because progress is measured by how safe people are and how inclusive systems are, not by whether a certain regime stays in power, it reframes incentives to serve citizens. **Trust-restorative:** as people see corruption being punished (or at least curbed) and experience far less violence (no war in their region, less crime in streets), their trust in authorities and the social contract rebounds. Legal identity fosters trust too: people feel seen by their government and entitled to services/justice. An independent judiciary visibly correcting wrongs or addressing grievances reassures people that rule of law exists (countering the fatalism that the powerful always get away). Achieving these would mean countries are dramatically safer and cleaner in governance, directly answering some of citizens' biggest demands, thereby validating institutional legitimacy.

17. **Partnerships for the Goals (Reconstructed):** *“Revitalize global partnership by meeting 100% of ODA (0.7% GNI) commitments, mobilizing \$500B/year in South-South and private SDG finance, and establishing an equitable decision-making reform in IFIs by 2025, monitored by a multilateral partnership accountability panel.”*

Why & How: This sets numeric targets for means: rich countries deliver on aid promises fully (0.7% GNI – a long-standing yet unfulfilled vow ²⁰). Also quantify South-South and private finance flows (\$500B/year is illustrative, to galvanize beyond aid). And crucially a reform of international financial institutions (IFIs) governance by 2025 to give developing countries more say – an outcome many have sought. These are bold and **outcome-bound** in the sense of concrete deliverables. **Symmetry-preserving:** equitable decision-making in IFIs (like World Bank/IMF voting shares, etc.) directly addresses power imbalances. Meeting ODA commitments and boosting South-South cooperation provides resources to those who need, balancing development gaps. **Adversarially auditable:** a multilateral partnership accountability panel (maybe under ECOSOC or independent) would track these targets yearly – who paid what, where private flows went, whether IFI reforms happened – and publicly name those falling short. This peer pressure and transparency makes it hard for, say, a G7 country to quietly shirk aid increases. It's **failure-visible:** if by 2025 IFI reform is not done, that's a glaring public failure; if money flows are short, the panel reports a financing gap and identifies culprits. Private sector promises (like philanthropic or impact investment pledges) would also be tracked, to hold them accountable beyond press releases. **Capture-minimizing:** ironically, by giving more voice to poorer countries in global bodies, it reduces capture of global rules by a few rich states or corporations. The accountability panel being multilateral (including various stakeholders) ensures no single bloc controls the narrative – it might include civil society to call out greenwashing. Also, having clear numeric obligations (0.7% aid, etc.) limits the ability to recast existing efforts as “new partnership” – they must genuinely add finance. **Trust-restorative:** meeting long-broken promises like 0.7% aid would significantly rebuild trust with developing countries – it shows good faith. IFI reform would be a historic step toward a more democratic global order, which could ease resentment and increase buy-in to global initiatives. For citizens, seeing their country either receive promised support or contribute fairly (in case of richer citizens, knowing their nation is doing its part) bolsters faith in international cooperation. An effective partnership framework convinces all parties that collaboration yields results and that big players aren't just dominating – which is essential for collective action on all other goals. Essentially, it says “we are all in this together and here's the proof: money on the table and shared decision power,” which is the foundation of credibility for the SDGs.

These reconstructed goals, while ambitious, are designed to **align incentives with genuine progress**, prevent easy outs or cosmetic compliance, and embed accountability at every level. They represent a shift from purely aspirational phrasing to **contractual targets** between leaders and citizens (and among nations) – targets that adversaries would find harder to subvert without immediate detection and consequence.

Retirement and Kill-Switch Criteria for Goals

No strategy is foolproof. Even reengineered goals may encounter unforeseen failure modes or adverse effects. Thus, we propose a **“retirement/kill-switch” protocol**: criteria and processes to modify or sunset a goal if it’s persistently compromised or counterproductive. This ensures the SDG framework can self-correct rather than perpetuate broken goals. Key elements include:

- **Regular Stress Tests:** Every goal should undergo a formal independent review at mid-point (e.g., 5-year intervals) assessing whether implementation is yielding results or being gamed. If a review (by say, the UN SDG Panel or an external auditor coalition) finds systematic issues – e.g., data manipulation, perverse outcomes, or stagnation – it can trigger a “goal at risk” status.
- **Pre-defined Kill Criteria:** For each goal, we define quantitative red lines. For example: if global hunger (Goal 2) increases for three consecutive years despite efforts, or if emissions (Goal 13) are not on a trajectory (<X Gt by year Y) for 1.5°C by 2025, these would meet criteria that the current approach is failing. Other criteria could be public trust metrics (surveys) dropping below a threshold in the goal area, or key input commitments (like finance for Goal 17) not materializing by a deadline. These objective triggers avoid ambiguous debates – it’s a fail-safe mechanism based on data.
- **Sunset or Revise Decision:** Upon trigger, an emergency SDG summit would convene stakeholders to decide: either **retire the goal** (if deemed unachievable or harmful in current form) or **radically reform** it (new targets, different implementation approach, possibly changing institutions responsible). Retirement is a last resort, essentially acknowledging “this goal as is does not work, and pursuing it further may waste resources or credibility.” A retired goal could be replaced by a more feasible or better-scoped goal if the overarching issue is still pertinent, or its components folded into other goals.
- **Mitigation Plans:** If a goal is retired, a mitigation plan addresses potential fallout. For instance, if Goal 8 (growth) was retired due to conflict with climate, a plan might focus on pursuing decent work under Goal 10 and Goal 13 synergy instead, so aspects are still covered. Retirement doesn’t mean abandoning the values, but admitting the framework needs change. Communication to the public would stress that ending a goal is to avoid doing harm or being dishonest, thus **maintaining trust by being frank**.
- **Institutional Kill-Switch Authority:** We recommend establishing a high-level “Goal Integrity Council” – perhaps composed of members from the UN, sovereign watchdog agencies, and citizen representatives (maybe from the red team coalition and frontier labs as per the prompt’s stakeholders). This council would have the mandate to recommend goal termination or overhaul when evidence compels, insulating the process from political reluctance. It acts somewhat like a central bank for SDGs – making tough calls for long-term stability of the agenda.
- **Examples:** Using current trends, one might foresee kill-switch use on, say, Goal 8 if pursuing GDP growth universally clearly undermines climate goals – better to officially drop or redefine it

than pretend. Or Goal 10, if inequality keeps worsening by 2027 despite efforts, the council might demand a new strategy (or admit that without stronger global tax cooperation, the goal can't advance under current conditions, forcing focus on that systemic change as a prerequisite). The criteria should be set now, not ad hoc later, to avoid bias – e.g., “if global Gini or other inequality index hasn't improved by X by year Y, Goal 10 is considered ineffective and up for re-mandate.”

The kill-switch concept is admittedly radical in international politics (where goals are usually added, not removed). But it introduces a much-needed **adaptive capacity**. It prevents the sink of resources into failing strategies and shows stakeholders that the framework values **results over vanity**. By willing to trim its own agenda in light of evidence, the UN and partners demonstrate integrity, which ultimately can **increase trust** – stakeholders see that goals aren't kept for political face if they don't work. Instead, the focus stays on *actual problem-solving*.

Importantly, the threat of a kill-switch can incentivize better performance. Knowing that a goal might be publicly retired (an embarrassment for those championing it) if they don't shape up could motivate actors to earnestly implement changes now. It's akin to a form of accountability pressure on goal sponsors.

In sum, the retirement/kill-switch criteria act as the system's emergency brake and self-pruning mechanism, ensuring the SDGs remain **flexible, honest, and focused**. They align with the red team ethos: be ready to sacrifice even cherished goals if they become liabilities, and do so systematically, not arbitrarily.

Meta-Risk Analysis

Even with the above improvements, we must acknowledge and plan for **meta-risks** – higher-level risks that affect the entire SDG framework or arise from the interplay of goals and the adversarial context:

- **Institutional Co-optation Risk:** There's a risk that even a fortified SDG framework could be co-opted by great power competition or corporate influence at the systemic level. For instance, one bloc of countries might selectively support certain goals that align with their strategic narrative (e.g., pushing infrastructure investments as rivalry tools, or using anti-corruption drives to delegitimize adversaries selectively). This could distort resource allocation and global cooperation. Mitigation: keep goal oversight bodies as multipolar and multi-stakeholder as possible (involving non-state voices to counterbalance state rivalries), and foster a norm that SDGs are neutral ground – an attack on one is an attack on the shared Earth, not an advantage to an opponent. Also ensure transparency of funding sources for SDG programs so that undue influence is spotted.
- **Trade-off and Perverse Coupling Risk:** Goals can conflict – e.g., aggressive climate action (Goal 13) could, if mishandled, increase short-term inequality (Goal 10) via energy prices; expanding industry (Goal 9) can clash with life on land (Goal 15) if not green. A meta-risk is that solving one goal in isolation exacerbates another, undermining overall success (a classic example cited in literature: biofuel push for climate harming food security ¹⁷). Our reconstruction tries to embed constraints to avoid this, but unforeseen trade-offs will arise. Mitigation: adopt a **“do no harm” test for inter-goal impacts** when designing policies – e.g., before a country implements a plan to achieve one goal, evaluate its effects on others (using tools like the SDG Interaction Framework ²⁵). Strengthen integrated planning units globally and nationally that

oversee cross-goal consistency (the original UN idea of indivisibility of SDGs needs real operational teeth).

- **Data and Modeling Risk:** Red-teaming largely assumes we can measure and detect issues. But if data systems fail (due to lack of capacity, cyber-attacks falsifying data, or novel types of progress that elude current metrics), the whole adversarial auditing falls apart. We see hints: reliance on big data/AI might introduce biases or manipulation (bad actors flooding false data). And complex models (for climate, economy) carry uncertainty – a meta-risk is betting on a strategy that models say will work, but reality diverges (model risk). Mitigation: invest in independent, redundant monitoring systems (satellite, IoT sensors, citizen science) to cross-verify official data ²⁶. Also incorporate **forecasting tournaments and scenario simulations** as part of SDG management – constantly test whether our interventions are yielding expected indicators, and if not, question our models and assumptions openly. Essentially, apply red team approaches continuously on the data/model side too.
- **Political Backlash and Legitimacy Risk:** Pushing a stringent, audit-heavy global agenda may itself provoke backlash if seen as infringing on sovereignty or cultural norms. For example, some regimes might lash out that SDGs have become “policing tools” of the UN or foreign entities – framing our improved accountability as neo-colonial oversight. Domestically, certain policies (like fuel subsidy cuts or anti-corruption crackdowns) could trigger unrest, which populists might use to discredit the SDGs (“IMF riots” phenomenon generalized to SDGs). This risk is paradoxical: better enforcement can feed narratives of external control. Mitigation: emphasize local ownership – e.g., national red teams so critique comes from within, not just outside. Use regional peer accountability (African Union, ASEAN etc. doing reviews) to avoid one-size-fits-all enforcement image. And communicate success stories loudly: as goals deliver visible benefits, public opinion can swing in favor, disarming backlash. We must pair tougher standards with **diplomacy and narrative framing** that these are tools to help governments help their people, not punish them.
- **Resource and Attention Fatigue Risk:** The comprehensive nature of 17 goals is already taxing; adding rigorous oversight and needing simultaneous leaps (like 50% emissions cut, 0.7% aid, etc.) demands immense political will and resources. A meta-risk is **agenda overload or donor fatigue** – actors simply cannot or will not sustain effort across all fronts, especially if crises (pandemics, wars) distract. If many goals are underachieved mid-way, there could be a collective loss of morale (“SDG fatigue”), causing even well-meaning stakeholders to disengage – a self-fulfilling failure. Mitigation: prioritize and sequence intelligently – our analysis suggests some goals are foundational (Goal 16, 17 for trust, finance; climate for planetary boundary) – so front-load those, and perhaps accept a triage approach if needed (not abandoning any, but focusing political capital on a few at a time). Also celebrate interim wins to keep momentum. Another tactic: modularize – break SDGs into manageable initiatives that stakeholders can tackle like pieces of a puzzle, scoring victories regularly rather than all-or-nothing in 2030.
- **Emergent Meta-Crisis Risk:** A scenario where multiple global stresses compound (e.g., climate disasters, financial crisis, a pandemic resurgence, geopolitical conflict) – could overwhelm the SDG system. In such a meta-crisis, even the best frameworks might falter as priorities shift to emergency response, and adversarial behavior spikes (resource nationalism, conflict). This is beyond SDG scope but affects it deeply. Mitigation: incorporate resilience into the framework – ensure that achievement of goals also builds capacity to handle shocks (e.g., climate adaptation, economic buffers, strong health systems – which are goals themselves). Have contingency protocols: if a crisis hits, how do we temporarily adapt targets or extend deadlines without abandoning principles? Perhaps a “crisis SDG solidarity clause” where countries agree not to roll

back certain cooperative measures even under duress (like not cutting aid or not weakening environmental rules for short-term gains).

- **Alignment with Frontier Risks:** The mention of frontier alignment labs in the prompt suggests AI or other advanced tech risk. A meta-risk is that the SDG approach doesn't keep up with rapidly evolving global threats (AI misalignment, biotech, cyber). If such a risk materializes (e.g., unchecked AI exacerbates inequality, or undermines information trust, or a biotech accident threatens health), it could render some goals moot or introduce new necessary goals. Mitigation: build a horizon-scanning function into the SDG apparatus (perhaps allied with those frontier labs) to update goals or add sub-goals addressing emerging issues. Ensure flexibility (the kill-switch principle but also birth-switch for new goals if needed). For example, if AI causes massive labor disruption, Goals 4 and 8 may need rewriting around AI governance. Being proactive on these meta-technological risks is key so they don't unravel progress.

In summary, the meta-risk landscape requires that our red-teamed SDG framework itself be **dynamic, self-critical, and resilient**. We have integrated some of that (kill-switch, etc.), but continuous foresight and adaptation will be needed. The worst outcome would be winning battles on individual goals but losing on these systemic fronts – that would degrade trust even more (“we did everything and still failed because of X”). Therefore, a meta-risk management strategy should be instituted, possibly overseen by an independent council (like the Goal Integrity Council which can also handle meta-risk strategy). It would identify cross-cutting dangers early and recommend coordinated adjustments across goals.

Ultimately, addressing meta-risks is part of making the SDG initiative itself **future-proof and hostile-resistant**. By foreseeing the above challenges and embedding mitigations, we aim to ensure that the SDG framework not only stands firm against known adversarial issues but is also agile in the face of new tests. This meta-layer of security and flexibility will be crucial for the SDGs to remain credible and effective through 2030 and beyond.

Open Questions and Limitations

Our comprehensive red team analysis and reconstruction of the SDGs closes many vulnerability gaps, but it also raises new questions and acknowledges certain limitations. In the spirit of institutional rigor and continuous improvement, we outline these openly:

- **Feasibility vs. Ambition:** Are the reconstructed goals realistically attainable, or do they set the bar so high (e.g., 50% emissions cut in a few years, zero poverty, etc.) that they risk widespread failure? We chose ambitious targets deliberately to galvanize action and avoid the trap of under-aiming (which invites complacency and capture) ³. However, there's an open question whether pushing such aggressive timelines could backfire politically or economically. For instance, will countries buy into these revisions, or will they disengage if targets seem out of reach? This pits ideal adversarial robustness against practical engagement. It may be that some goals need calibrated interim milestones (e.g., a 2025 checkpoint target) to keep efforts on track and believable.
- **Resource Allocation and Trade-offs:** While we attempted to ensure synergy, implementing all reconstructed goals concurrently will demand enormous resources and coordination. Even with increased global finance (Goal 17 reforms), there's a question of absorptive capacity: do countries have the institutional bandwidth to simultaneously overhaul governance, infrastructure, social policy, etc., at the pace demanded? One limitation is human and technical capacity – many nations will need unprecedented support to meet, say, data transparency or

independent audit requirements. Without addressing capacity building thoroughly (something often promised but under-delivered), the best goals can falter. How to scale up capacity quickly and ethically remains a question.

- **Measurement and Verification Challenges:** We have leaned heavily on independent monitoring (data, audits, satellites). However, this presumes a level of access and neutrality that might not always exist. Authoritarian states might block inspectors or censor data; companies might find new ways to obfuscate supply chains beyond current tech's ability to trace. Our plan to use cutting-edge tools (satellite, AI analytics) has limitations if adversaries invest in countermeasures (e.g., spoofing data, cyber-attacks on monitoring infrastructure). There is an ongoing question of who guards the guardians – ensuring the independence and integrity of the very auditors and systems we rely on ²⁶. We propose international bodies, but they too need governance to avoid being co-opted. The details of establishing truly independent, incorruptible verification mechanisms are complex and need further work.
- **Political Will and Sovereignty:** Ultimately, the SDGs are voluntary. Even reconstructed, enforcement relies on moral suasion, peer pressure, and domestic accountability. One must ask: will nations agree to the intrusiveness of these proposals (e.g., foreign satellites policing their fishing, global panels calling out their corruption)? Some may see it as an erosion of sovereignty. We've framed these in a way to appeal to mutual benefit and trust restoration, but the question stands whether a new global social contract can be forged to permit these incursions for the greater good. If not, we face a limitation that the adversarial framework is only as strong as the weakest link – a few large countries opting out could undermine global targets (e.g., climate) substantially. Diplomatic strategy around this is crucial and beyond our report's scope, but remains an open challenge.
- **Cultural and Ethical Considerations:** Avoiding ideological framing was a guideline, but inevitably some of these goals (like gender equality, governance reforms) do touch cultural norms and values. There's an open question: how to implement them in culturally diverse contexts without triggering backlash? For example, pushing for gender parity or legal identity in communities with different social structures requires sensitivity. Our analysis treated these as technical/adversarial problems, but the human element – winning hearts and minds – is a softer side we haven't solved here. The limitation is that red teaming can fix structural flaws, but not everything is structural; social change requires legitimacy and buy-in which can't be manufactured by audits alone. Further interdisciplinary work with anthropologists, communicators, local leaders would be needed.
- **Unintended Consequences:** We have tried to anticipate perverse incentives, but by creating hard targets we might inadvertently create new ones. For example, aiming to reduce violent deaths by 75% might tempt some regimes to under-report murders or reclassify political killings as "accidents" to improve stats – a new kind of manipulation. Or pushing renewables to 80% might lead to rushed deployment without proper planning, causing grid issues or raw material shortages. Each quantitative target should be stress-tested for potential side-effects. It's an open question if we've identified all major ones or if new gaming methods will arise. The kill-switch is a remedy for when things go wrong, but better would be ongoing adaptive management to tweak incentives before reaching kill-switch points.
- **Inter-goal Coordination:** We note that even reworked, goals like climate, inequality, and consumption are deeply interlinked; solving one greatly eases solving others, and failing one hampers others. Our framework does better at integrating (like environmental limits in Goal 8, etc.), but the governance of implementing these as a package is an unanswered practical

question. Do we expect each country to devise an integrated national strategy (possibly with red team units domestically)? How to ensure that ministries and sectors cooperate rather than work in silos on “their” goal? The UN could push integrated approaches, but that requires capacity too. So a limitation is not in concept but in execution: making the holistic approach operational on the ground. Further work could be developing country-level pilots of integrated SDG task forces.

- **Security and Retaliation Risks:** As we tighten oversight, there could be pushback not just politically but in cyber or physical terms. For instance, satellite monitoring could be jammed; auditors could face threats in corrupt countries; civil society monitors could be targeted. We need to consider safety protocols – a sad reality that red team measures might endanger whistleblowers or inspectors. This is an open ethical question: how to protect those who enforce accountability in hostile environments. Solutions might include diplomatic protections, funding for safe harbor programs, or technology that anonymizes data sources (so regimes can’t pinpoint who provided evidence). We acknowledge this as a serious limitation if not addressed – the system is only as good as the safety of its truth-tellers.
- **Timeline and Flexibility:** The 2030 deadline looms. Some of our proposals (like IFI reform by 2025) are extremely tight. If these are not done swiftly, is there flexibility to extend the timeline? Open question: would extending timelines (to 2035 or 2040 for some targets) harm momentum or is it necessary to be realistic? We didn’t explicitly cover timeline change, focusing on urgency, but realistically some goals might need more time. This should be debated openly rather than clandestinely adjusting expectations. Possibly a tiered timeline: critical actions by 2025 (foundation laid), majority targets by 2030, and final deep outcomes by 2040. The limitation is human institutions and behavior change often move slower than our targets require, so we must reconcile that or face disillusionment.
- **Role of Frontier Technologies:** While we touched on data and AI for monitoring, a big open question is how to responsibly harness new tech (AI, biotech, blockchain, etc.) to accelerate SDGs without falling prey to new risks. For example, blockchain could improve transparency (Goal 16, 17), AI could optimize resource use (Goal 12), but they also bring risks of bias or misuse. We haven’t deeply dived into this, but it’s both an opportunity and a challenge. Partnerships with frontier alignment labs (as mentioned in prompt) might yield innovative solutions (like AI systems monitoring SDG progress or forecasting interventions), but also require caution (ensuring AI decisions don’t encode biases that create new inequities – ironically violating Goal 10). This is flagged as an area for ongoing research and interdisciplinary collaboration.

In concluding, our red team report is **not a panacea**; it’s a strategic blueprint that must be iteratively refined. We have improved the SDG framework’s armor and swords, but it will still be wielded by human institutions in a complex world. The open questions above should be taken as a research and action agenda for the next phase: testing these recommendations in pilot programs, consulting diverse stakeholders for buy-in, and adjusting course as we learn.

Our work stands as a reference point – a rigorous attempt to envision a more resilient and honest global goal system. It will withstand scrutiny best if it is treated not as dogma but as a living document to be debated and updated. By acknowledging the uncertainties and limitations, we invite that very scrutiny and collaboration from labs, regulators, and institutions, aligning with the final aim: **to collectively outsmart and outpace the adversarial forces that threaten sustainable development, and to do so in a spirit of transparency and shared commitment.**

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